MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **CALCULATION**
SECONDARY CRUSHING
DYNAMIC ANALYSIS OF MODIFICATION OF CRUSHER SUPPORT STRUCTURE

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PREFACE

VIBRATION PROBLEM - SECONDARY CRUSHER & SUPPORTING STRUCTURE

A. The Problem:

- Crusher was started with VFD. At 50 % speed of crusher, the crusher assembly was vibrating beyond the expected limits. At 100% speed, the vibrations increased.
- Though vibration isolators have been provided below the Crusher, the vibration was being transmitted to the supporting concrete structure and could be felt by all.
- Vibration measurement was taken at 100% speed of crusher and maximum observed value in horizontal direction was 19.3 mm/s (RMS) on concrete deck support. ISO 10816 specifies an allowable limit of 4-5 mm/s, and the project specification limit is 3 mm/s. Manufacturer has not specified any allowable limit of vibration.

B. Root Cause Analysis

- For the applied pattern of dynamic loads, the maximum velocity at concrete support was computed as 4.46 mm/s. However, the actual measured velocity at concrete support was found to be 19.3 mm/s**.** This indicates that the load pattern considered in the analysis does not match the actual load arising from the operating equipment.
- Misinterpretation of the forced vibration nature of the equipment.
- Incompliance with project requirements with regard to being away from the range of excitation frequencies and vibration amplitude limit.
- Misinterpretation of the geotechnical advice with regard to foundation design parameters for dynamic foundations.
- A revised analysis has been done with a modified load pattern, as confirmed by the *equipment manufacturer. The modified loading pattern is presented in Chapter 3 and the* confirmation by the equipment manufacturer, is given in Appendix 2. A detailed soil strata modelling has also been done in this analysis.
- With the above modifications in analysis methodology, the computed velocity is 20mm/s, which matches the actual findings at site. However, this is much more than the allowable, comfortable vibration limit of 3mm/s.
- This is the reason why extreme vibration is being felt on the structure.

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C. Proposed Stiffening of Structure:

- To reduce the vibration level to an allowable limit, additional stiffening of the structure is required, with additional concrete walls, as shown in Appendix 1.
- The modified structure has been analysed with the corrected load pattern and with detailed soil strata modelling.

D. Conclusion:

- The stiffening has reduced the computed vibration to 2.3 mm/s, which is within the acceptable limit.
- Also, as a result of stiffening of the structure, the natural frequency is now 9.5 Hz which is 2.04 times the operating frequency of the equipment.
- The modified structure is adequate for the loads arising from the equipment.

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List of Contents **Pages**

1. Introduction 5
2. Design Basis 5
   - 2.1. Geometry of the Model 5
   - 2.2. Material Properties of Concrete 5
   - 2.3. Soil Properties 6
   - 2.4. Damping Property 13
   - 2.5. Lump Mass Modelling of Crusher 13
3. Modelling of Dynamic Loading 19
   - 3.1. Dynamic Load, normal operation (f=4.67 Hz) 19
   - 3.2. Philosophy of the Operational Dynamic Loadings 20
   - 3.3. Operational Loading in ANSYS 23
   - 3.4. Dynamic Load, Startup & Shutdown 25
   - 3.5. Boundary Conditions 26
4. Analysis of the Existing Structure 26
   - 4.1. Complete Model 26
   - 4.2. Mesh 27
   - 4.3. Harmonic Analysis 28
5. Analysis of the Proposed Structure 29
   - 5.1. Individual Component of the Model 30
   - 5.2. Complete Model 32
   - 5.3. Mesh 34
   - 5.4. Modal Analysis 35
   - 5.5. Harmonic Analysis 36
6. Conclusions 44
7. References 44
8. Appendix 1 45
9. Appendix 2 48
10. Comment Resolution Sheet 51

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1. Introduction

This report is a verification of the Secondary Crusher system, using finite element analysis [1]. The main purpose of this analysis is to investigate:

- Root cause analysis for the excessive vibration of the existing structure. Harmonic analyses are performed considering empty conditions of the crusher.
- Undamped modal behavior to check with the excitation frequencies of the proposed Secondary Crusher. Two modal analyses are performed considering empty and loaded conditions of the crusher.
- Steady state dynamic behavior of the proposed system with harmonic analysis to ensure that potential excitation forces will not cause vibration that might jeopardize safe operation of the equipment. Two harmonic analyses are performed considering empty and loaded conditions of the crusher.
- The concrete design on modifications is shown in revised Secondary Crushing Foundations and Superstructures (MD-216-3120-EG-CV-CAL-1004).

2. Design Basis

2.1. Geometry of the Model

Secondary Crusher system consists of the Soils, RCC footing, columns, beams, ties, slab, and different pedestals.

2.1.1. Existing Concrete Structure

Dimension of the existing concrete structure is given in Appendix 1.

2.1.2. Proposed Concrete Structure

Dimension of the proposed concrete structure is given in Appendix 1.

2.2. Material Properties of Concrete

The material properties of concrete shown in Table 1a.

Property	Value	Unit
Material Field Variables	Table	
Density	2400	kg m^-3
Isotropic Secant Coefficient of Thermal Expansion		
Isotropic Elasticity		
Derive from	Young's Mod...	
Young's Modulus	3.1528E+10	Pa
Poisson's Ratio	0.17	
Bulk Modulus	1.5923E+10	Pa
Shear Modulus	1.3474E+10	Pa

Table 1a. Material properties of concrete

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2.3. Soil Properties

Borehole 19 is near Crusher System. Shear modulus & Poisson ratios are taken from the soil report to model the soil in different layers. The material properties of soil in Figure 1. Dimension of the soil model is considered as 45m X 45m X 15m depth.

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TEST REPORT - DOWNHOLE SEISMIC TEST RESULTS *(ASTM D7400 - 08)*

FORM # RGF-KHO-GEO-DH-1-00

Project: *MANSOURAH MASSARAH GOLD PROJECT*
Client: *L & T CONSTRUCTION*
Project No.: *19-K-3054* **Test No.** *DH-03*
Borehole No.: *BH-19* **Geotechnical File No.:** -
Source Type: *Sledge Hammer* **Source to BH Distance (m):** 3 m
Receiver Type: *Triaxial Geophones* **Coordinates (m):** *N: 2506510.84, E: 264424.58*

Sr. No.	Depth of Reciever (m)	Distance of Source to Receiver (m)	Average Time		Calculated Velocity		Assumed Density	Poisson Ratio	Shear Modulus	Young Modulus	Bulk Modulus
			P-wave (msec)	S-wave (msec)	P-wave (m/sec)	S-wave (m/sec)	ρ (g/cm³)	υ	G (M Pa)	E (M Pa)	K (M Pa)
1	0.75	3.09	5.10	9.09	606	340	1.7	0.27	197	500	363
2	1.50	3.35	5.38	9.56	935	557	1.7	0.22	527	1292	783
3	2.25	3.75	5.78	10.21	990	609	1.8	0.20	668	1596	873
4	3.00	4.24	6.28	11.01	985	616	1.8	0.18	683	1610	837
5	3.75	4.80	6.87	11.97	949	583	1.8	0.20	612	1464	804
6	4.50	5.41	7.48	12.93	993	631	1.8	0.16	717	1666	820
7	5.25	6.05	8.16	14.04	939	575	1.8	0.20	595	1428	793
8	6.00	6.71	8.81	15.09	1018	630	1.8	0.19	714	1699	912
9	6.75	7.39	9.44	16.15	1077	640	1.8	0.23	737	1809	1104
10	7.50	8.08	9.94	16.97	1382	843	1.9	0.20	1350	3250	1830
11	8.25	8.78	10.26	17.50	2190	1322	2.1	0.21	3671	8908	5176
12	9.00	9.49	10.59	18.04	2146	1312	2.2	0.20	3785	9099	5089
13	9.75	10.20	10.90	18.55	2304	1401	2.2	0.21	4315	10417	5926
14	10.50	10.92	11.22	19.08	2247	1357	2.2	0.21	4050	9825	5709
15	11.25	11.64	11.58	19.66	2008	1246	2.2	0.19	3418	8113	4315
16	12.00	12.37	11.90	20.18	2269	1397	2.2	0.20	4291	10256	5609
17	12.75	13.10	12.23	20.74	2209	1302	2.2	0.23	3727	9198	5763
18	13.50	13.83	12.56	21.26	2216	1406	2.2	0.16	4349	10115	5000
19	14.25	14.56	12.96	21.89	1833	1164	2.2	0.16	2979	6924	3417
20	15.00	15.30	13.42	22.63	1597	993	2.2	0.19	2169	5140	2721
21	15.75	16.03	13.87	23.39	1636	969	2.2	0.23	2064	5077	3135
22	16.50	16.77	14.26	24.02	1891	1170	2.2	0.19	3014	7168	3846

Figure 1a. Soil Properties for different layers

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Properties of Outline Row 11: Rock1

	A	B	
1	Property	Value	
2	Material Field Variables	Table	
3	Density	1700	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Shear Modulus and Poisson's...	
8	Young's Modulus	5.0038E+08	Pa
9	Poisson's Ratio	0.27	
10	Bulk Modulus	3.6259E+08	Pa
11	Shear Modulus	1.97E+08	Pa

Properties of Outline Row 22: Rock2

	A	B	
1	Property	Value	
2	Material Field Variables	Table	
3	Density	1700	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Shear Modulus and Poisson's...	
8	Young's Modulus	1.2859E+09	Pa
9	Poisson's Ratio	0.22	
10	Bulk Modulus	7.654E+08	Pa
11	Shear Modulus	5.27E+08	Pa

Properties of Outline Row 26: Rock3

	A	B	
1	Property	Value	
2	Material Field Variables	Table	
3	Density	1800	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Shear Modulus and Poisson's...	
8	Young's Modulus	1.6032E+09	Pa
9	Poisson's Ratio	0.2	
10	Bulk Modulus	8.9067E+08	Pa
11	Shear Modulus	6.68E+08	Pa

Properties of Outline Row 27: Rock4

	A	B	
1	Property	Value	
2	Material Field Variables	Table	
3	Density	1800	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Shear Modulus and Poisson's...	
8	Young's Modulus	1.6119E+09	Pa
9	Poisson's Ratio	0.18	
10	Bulk Modulus	8.3952E+08	Pa
11	Shear Modulus	6.83E+08	Pa

Figure 1b. Properties of soil layers 1 to 4

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Properties of Outline Row 28: Rock5

	A	B	
1	Property	Value	
2	Material Field Variables	Table	
3	Density	1800	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Shear Modulus and Poisson's...	
8	Young's Modulus	1.4688E+09	Pa
9	Poisson's Ratio	0.2	
10	Bulk Modulus	8.16E+08	Pa
11	Shear Modulus	6.12E+08	Pa

Properties of Outline Row 29: Rock6

	A	B	
1	Property	Value	
2	Material Field Variables	Table	
3	Density	1800	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Shear Modulus and Poisson's...	
8	Young's Modulus	1.6634E+09	Pa
9	Poisson's Ratio	0.16	
10	Bulk Modulus	8.1541E+08	Pa
11	Shear Modulus	7.17E+08	Pa

Properties of Outline Row 30: Rock7

	A	B	
1	Property	Value	
2	Material Field Variables	Table	
3	Density	1800	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Shear Modulus and Poisson's...	
8	Young's Modulus	1.428E+09	Pa
9	Poisson's Ratio	0.2	
10	Bulk Modulus	7.9333E+08	Pa
11	Shear Modulus	5.95E+08	Pa

Properties of Outline Row 31: Rock8

	A	B	
1	Property	Value	
2	Material Field Variables	Table	
3	Density	1800	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Shear Modulus and Poisson's...	
8	Young's Modulus	1.6993E+09	Pa
9	Poisson's Ratio	0.19	
10	Bulk Modulus	9.1361E+08	Pa
11	Shear Modulus	7.14E+08	Pa

Figure 1c. Properties of soil layers 5 to 8

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Properties of Outline Row 32: Rock9

	A	B	
1	Property	Value	
2	Material Field Variables	Table	
3	Density	1800	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Shear Modulus and Poisson's...	
8	Young's Modulus	1.813E+09	Pa
9	Poisson's Ratio	0.23	
10	Bulk Modulus	1.1191E+09	Pa
11	Shear Modulus	7.37E+08	Pa

Properties of Outline Row 12: Rock10

	A	B	
1	Property	Value	
2	Material Field Variables	Table	
3	Density	1900	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Shear Modulus and Poisson's...	
8	Young's Modulus	3.24E+09	Pa
9	Poisson's Ratio	0.2	
10	Bulk Modulus	1.8E+09	Pa
11	Shear Modulus	1.35E+09	Pa

Properties of Outline Row 13: Rock11

	A	B	
1	Property	Value	
2	Material Field Variables	Table	
3	Density	2100	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Shear Modulus and Poisson's...	
8	Young's Modulus	8.8838E+09	Pa
9	Poisson's Ratio	0.21	
10	Bulk Modulus	5.1056E+09	Pa
11	Shear Modulus	3.671E+09	Pa

Properties of Outline Row 14: Rock12

	A	B	
1	Property	Value	
2	Material Field Variables	Table	
3	Density	2200	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Shear Modulus and Poisson's...	
8	Young's Modulus	9.084E+09	Pa
9	Poisson's Ratio	0.2	
10	Bulk Modulus	5.0467E+09	Pa
11	Shear Modulus	3.785E+09	Pa

Figure 1d. Properties of soil layers 9 to 12

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Properties of Outline Row 15: Rock13

	A	B	
1	Property	Value	
2	Material Field Variables	Table	
3	Density	2200	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Shear Modulus and Poisson's...	
8	Young's Modulus	1.0442E+10	Pa
9	Poisson's Ratio	0.21	
10	Bulk Modulus	6.0013E+09	Pa
11	Shear Modulus	4.315E+09	Pa

Properties of Outline Row 16: Rock14

	A	B	
1	Property	Value	
2	Material Field Variables	Table	
3	Density	2200	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Shear Modulus and Poisson's...	
8	Young's Modulus	9.801E+09	Pa
9	Poisson's Ratio	0.21	
10	Bulk Modulus	5.6328E+09	Pa
11	Shear Modulus	4.05E+09	Pa

Properties of Outline Row 17: Rock15

	A	B	
1	Property	Value	
2	Material Field Variables	Table	
3	Density	2200	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Shear Modulus and Poisson's...	
8	Young's Modulus	8.1348E+09	Pa
9	Poisson's Ratio	0.19	
10	Bulk Modulus	4.3736E+09	Pa
11	Shear Modulus	3.418E+09	Pa

Properties of Outline Row 18: Rock16

	A	B	
1	Property	Value	
2	Material Field Variables	Table	
3	Density	2200	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Shear Modulus and Poisson's...	
8	Young's Modulus	1.0298E+10	Pa
9	Poisson's Ratio	0.2	
10	Bulk Modulus	5.7213E+09	Pa
11	Shear Modulus	4.291E+09	Pa

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Figure 1e. Properties of soil layers 13 to 16

Properties of Outline Row 19: Rock17

	A	B	
1	Property	Value	
2	Material Field Variables	Table	
3	Density	2200	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Shear Modulus and Poisson's...	
8	Young's Modulus	9.1684E+09	Pa
9	Poisson's Ratio	0.23	
10	Bulk Modulus	5.6595E+09	Pa
11	Shear Modulus	3.727E+09	Pa

Properties of Outline Row 20: Rock18

	A	B	
1	Property	Value	
2	Material Field Variables	Table	
3	Density	2200	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Shear Modulus and Poisson's...	
8	Young's Modulus	1.009E+10	Pa
9	Poisson's Ratio	0.16	
10	Bulk Modulus	4.9459E+09	Pa
11	Shear Modulus	4.349E+09	Pa

Properties of Outline Row 21: Rock19

	A	B	
1	Property	Value	
2	Material Field Variables	Table	
3	Density	2200	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Shear Modulus and Poisson's...	
8	Young's Modulus	6.9113E+09	Pa
9	Poisson's Ratio	0.16	
10	Bulk Modulus	3.3879E+09	Pa
11	Shear Modulus	2.979E+09	Pa

Properties of Outline Row 23: Rock20

	A	B	
1	Property	Value	
2	Material Field Variables	Table	
3	Density	2200	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Shear Modulus and Poisson's...	
8	Young's Modulus	5.1622E+09	Pa
9	Poisson's Ratio	0.19	
10	Bulk Modulus	2.7754E+09	Pa
11	Shear Modulus	2.169E+09	Pa

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Figure 1f. Properties of soil layers 17 to 20

Properties of Outline Row 24: Rock21

	A	B	
1	Property	Value	
2	Material Field Variables	Table	
3	Density	2200	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Shear Modulus and Poisson's...	
8	Young's Modulus	5.0774E+09	Pa
9	Poisson's Ratio	0.23	
10	Bulk Modulus	3.1342E+09	Pa
11	Shear Modulus	2.064E+09	Pa

Properties of Outline Row 25: Rock22

	A	B	
1	Property	Value	
2	Material Field Variables	Table	
3	Density	2200	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Shear Modulus and Poisson's...	
8	Young's Modulus	7.1733E+09	Pa
9	Poisson's Ratio	0.19	
10	Bulk Modulus	3.8566E+09	Pa
11	Shear Modulus	3.014E+09	Pa

Figure 1g. Properties of soil layers 21 to 22

2.4. Damping Property

Damping of 5% is considered for dynamic analysis, shown in Table 2.

Details of "Analysis Settings"

Options	
Rotordynamics Controls	
Output Controls	
Damping Controls	
Constant Damping Ratio	5.e-002
Stiffness Coefficient Define By	Direct Input
Stiffness Coefficient	0.
Mass Coefficient	0.
Analysis Data Management	

Table 2. Damping properties for harmonic analysis

2.5. Lump Mass Modelling of Crusher

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Mass model of the crusher, listed in Table 2a & 2b, is considered as equivalent amount mass of material for simplification of the model in finite element analysis. Modelling is done taking information from load data given by equipment manufacturer shown A [1,2], B [1,2], and C [1,2] loading as follows:

Loaded Conditions:

- Mass of A1, A2, B1, & B2: 200kN = around 20,000 kg each
- Mass of C1, & C2: 45kN = around 4,500 kg each

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	Static Load	Dynamic Load, normal operation f=4,67 Hz	
Damper position	Vertical	Vertical	Horizontal
A [1, 2]	200 kN	± 13 kN	15 kN
B [1, 2]	200 kN	± 14 kN	16 kN
C [1, 2]	45 kN	± 4 kN	2 kN
Vertical load, positive downwards			
Horizontal load, (all horizontal directions)			

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Table 2a. Load data from [MD - 216 -3120 - EG- ME -GAD -V500-1001 Rev C01] [2]

Empty Conditions:

- Mass of A1, A2, B1, & B2: 145.75kN = around 14,575 kg each
- Mass of C1, & C2: 45kN = around 4,500 kg each

TOTAL WEIGHT	: 58,300 Kg
MAX LENGTH WITH SUBFRAME	: 6,350 mm
WIDTH WITH SUB FRAME	: 3200 mm
HEIGHT	: 4700 mm

Table 2b. Load data from [MD - 216 -3120 - EG- ME -GAD -V500-1001 Rev C01] [2]

The material properties of lumped structures and other components are considered to get such masses and listed in Figure 2a, 2b, & 2c.

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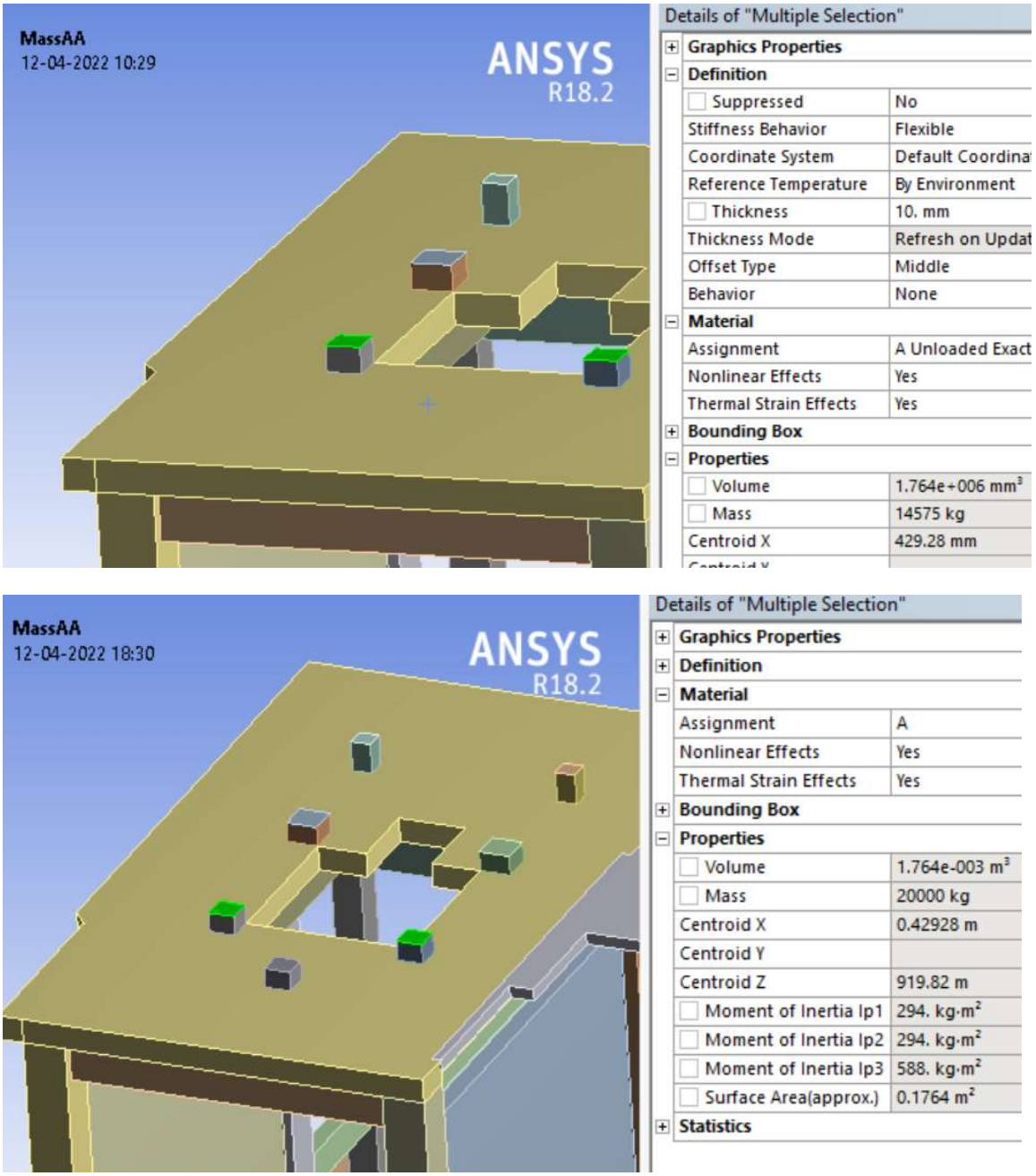
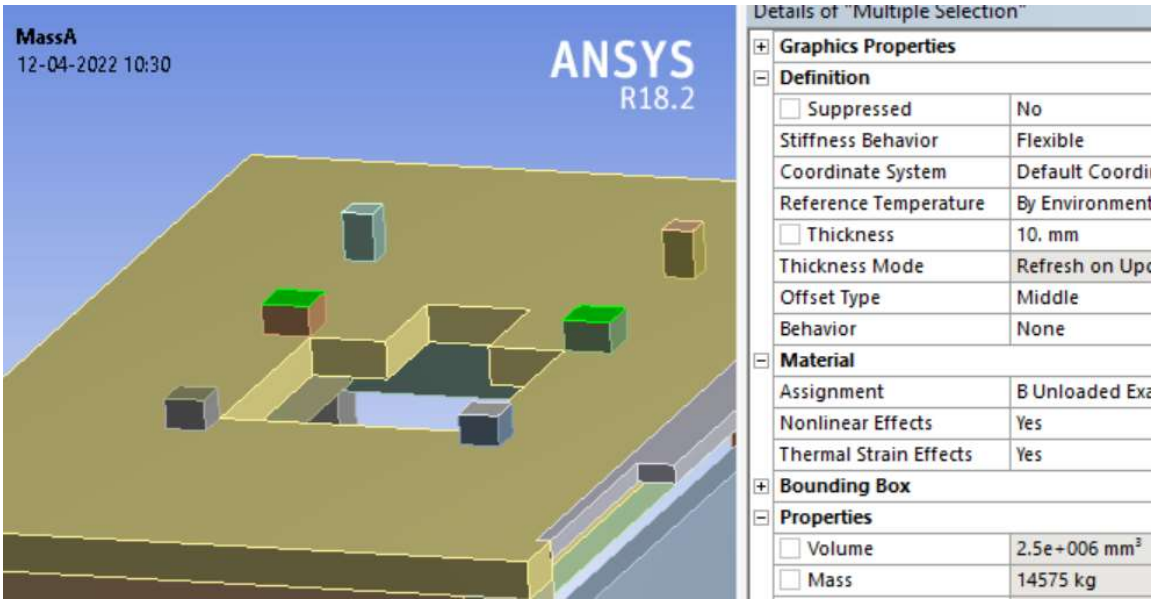


Figure 2a. Lumped mass of the crushers at support: Unloaded & Loaded

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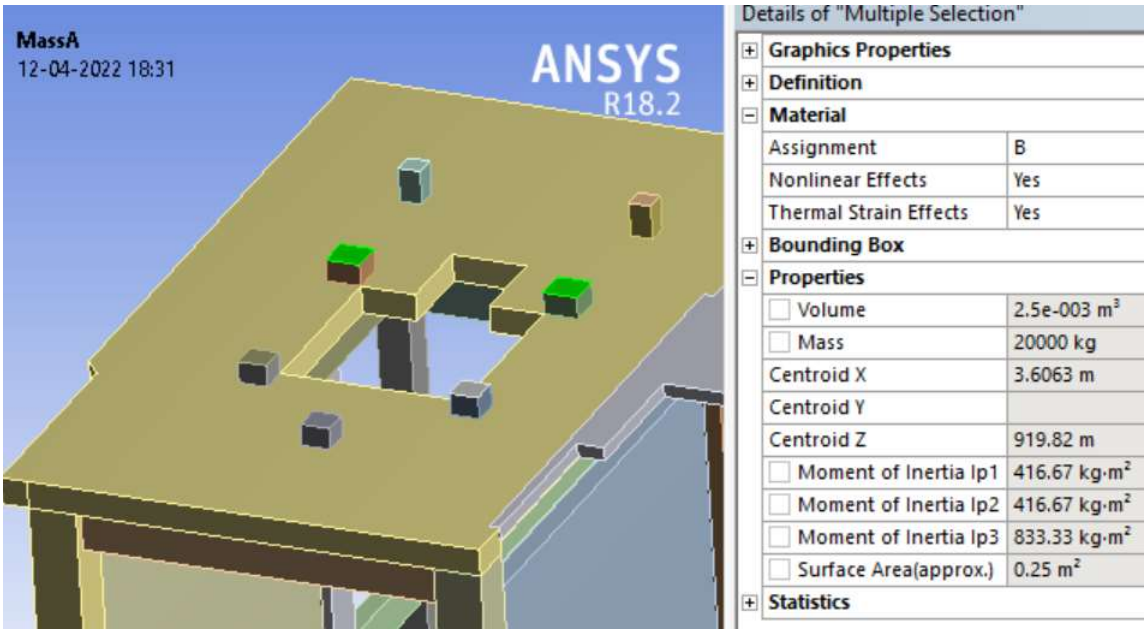


Figure 2b. Lumped mass of the crushers at support: Unloaded & Loaded

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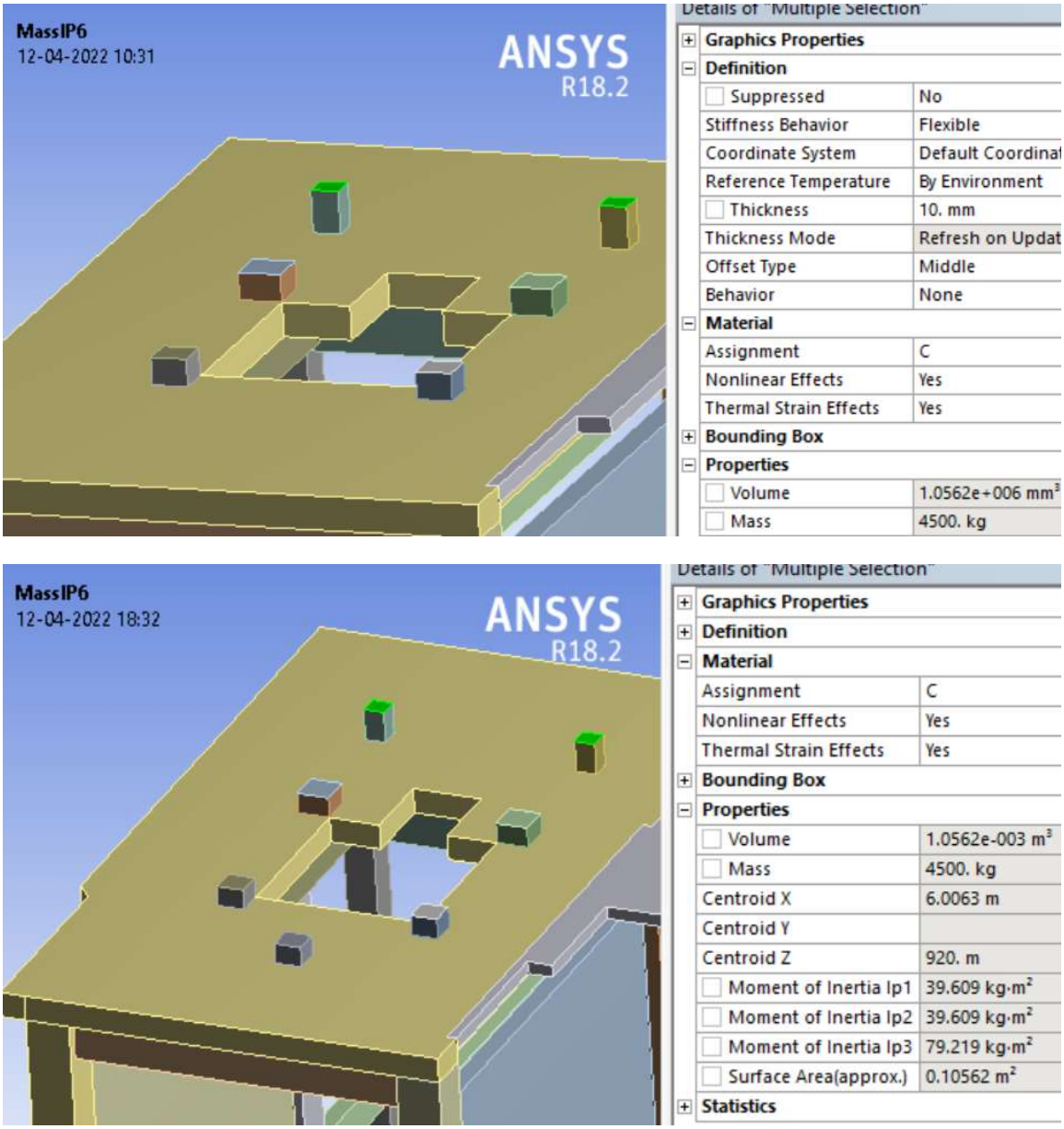


Figure 2c. Lumped mass of the crushers at support: Unloaded & Loaded

3. Modelling of Dynamic Loading

3.1. Dynamic Load, normal operation (f=4.67 Hz)

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Components of dynamic forces are taken from drawing provided by equipment manufacturer. Continuous component (Static load) has no effect in dynamic analysis, so ignored in current analysis. Dynamic component is applied in the respective pedestals of the system with different phases and frequency (4.67 Hz).

3.2. Philosophy of the Operational Dynamic Loadings

All horizontal forces are acting in the same horizontal direction in XY plane as shown in Figure 3a-d. Vertical loads are applied to increase the rocking effect in respective phases.

3.2.1. Time phase 0 Degree

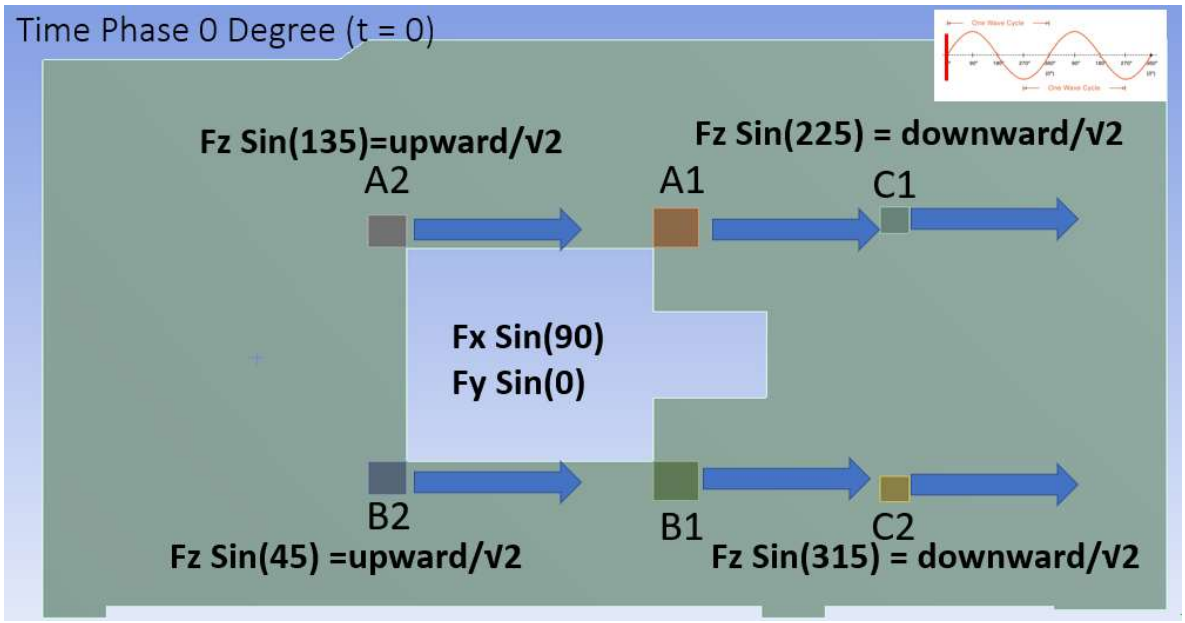


Figure 3a. Excitation Force Direction & Phase angles

3.2.2. Time phase 90 Degree

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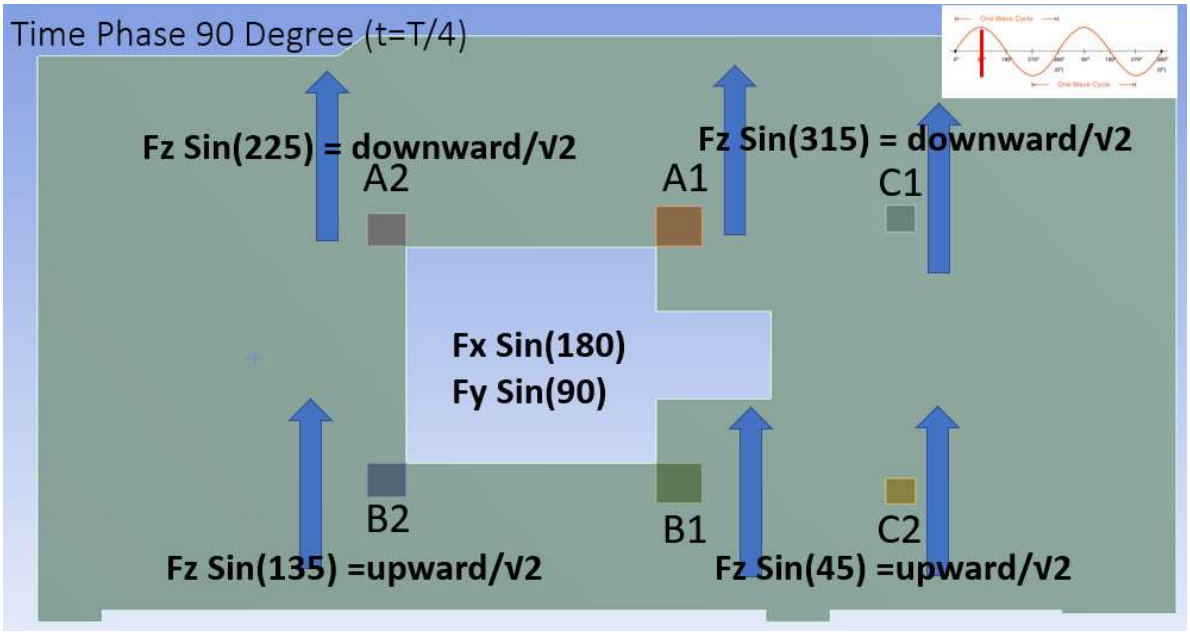


Figure 3b. Excitation Force Direction & Phase angles

3.2.3. Time phase 180 Degree

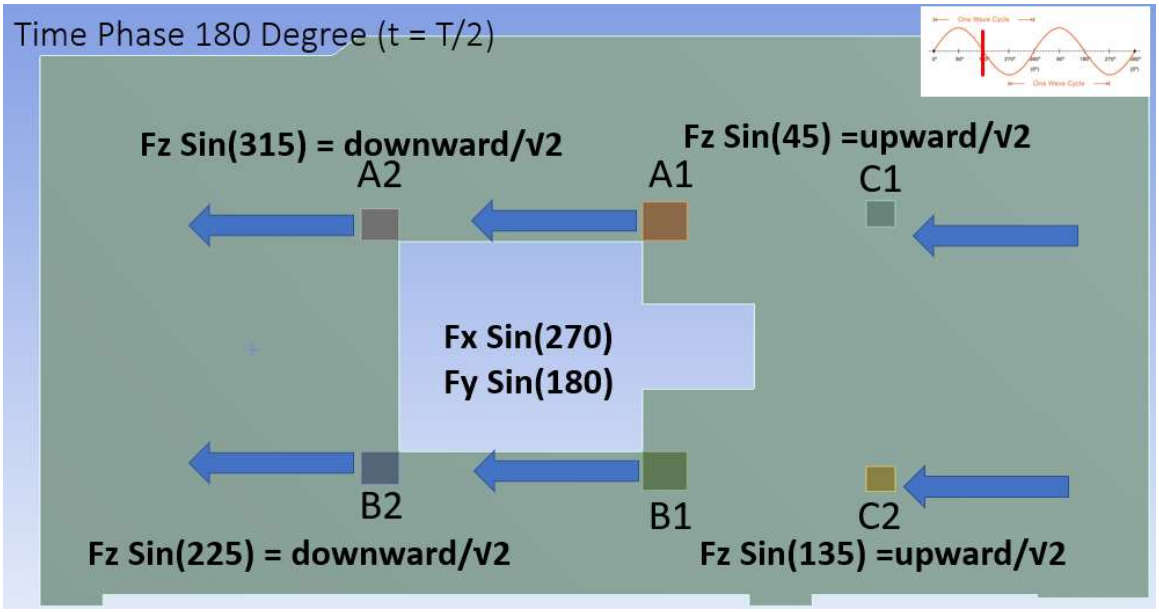


Figure 3c. Excitation Force Direction & Phase angles

3.2.4. Time phase 270 Degree

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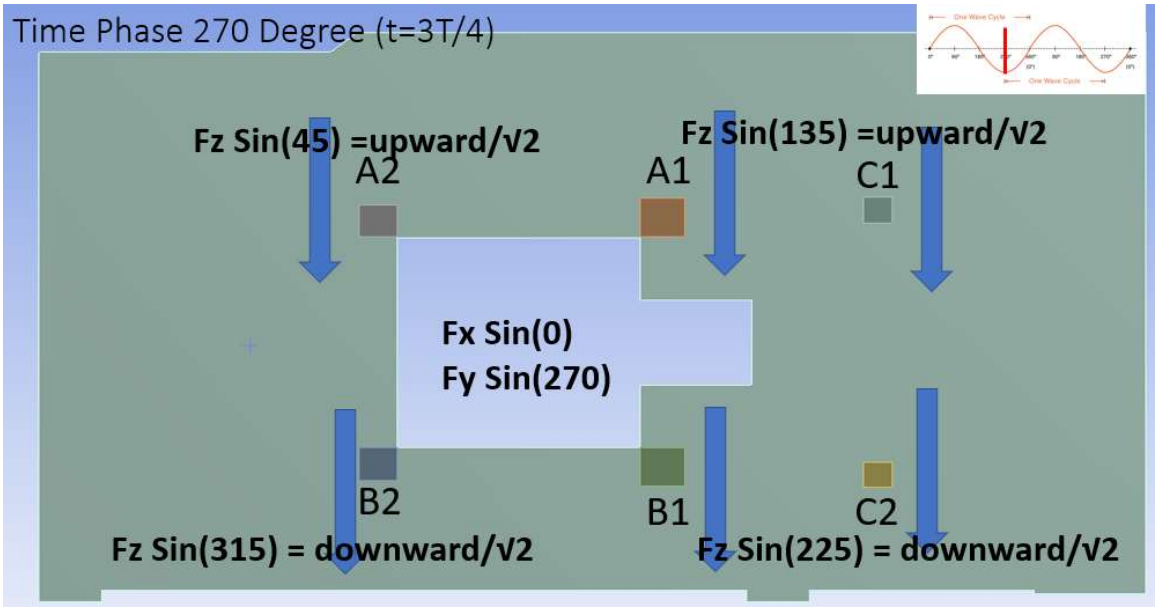


Figure 3d. Excitation Force Direction & Phase angles

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3.3. Operational Loading in ANSYS

3.3.1. Phases of the Loadings

Following phase angles are given in different pedestals tops:

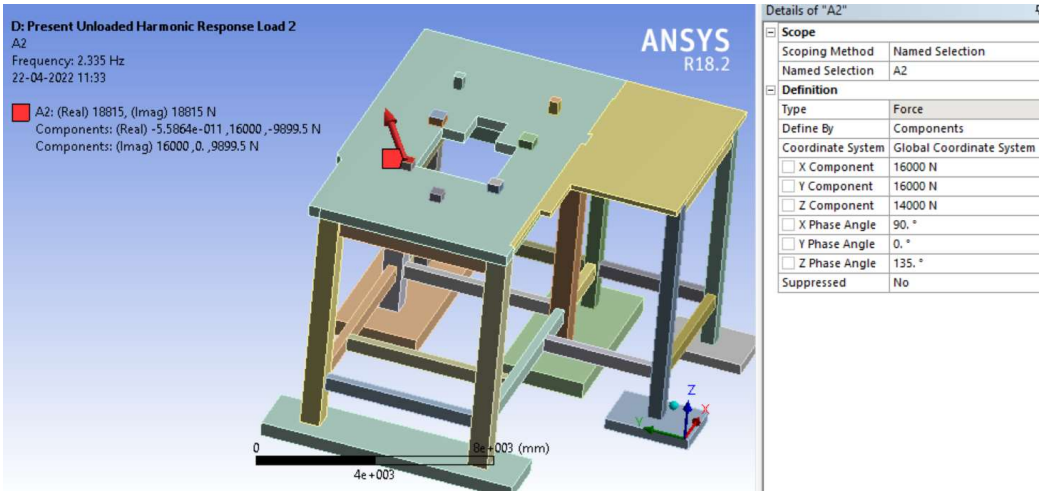


Figure 3e. Excitation Force & Phase angles

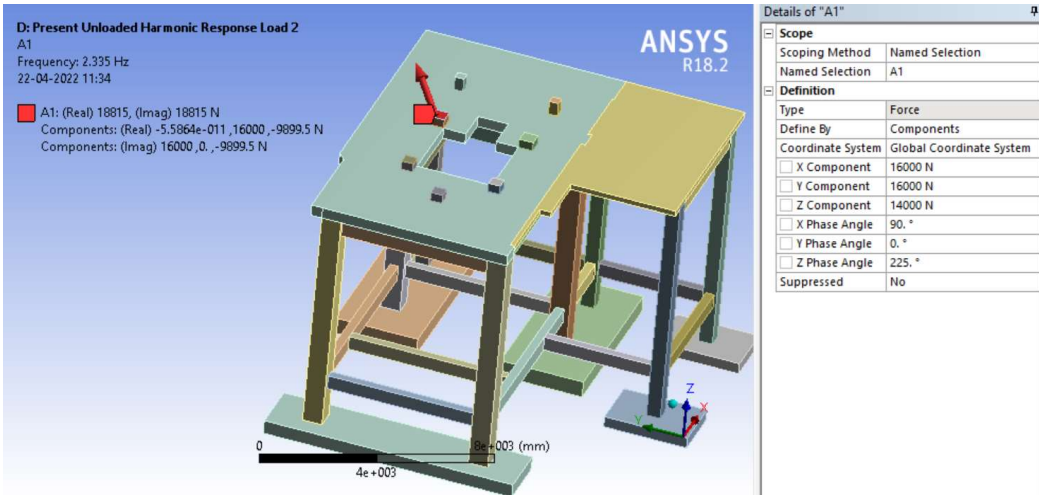


Figure 3f. Excitation Force & Phase angles

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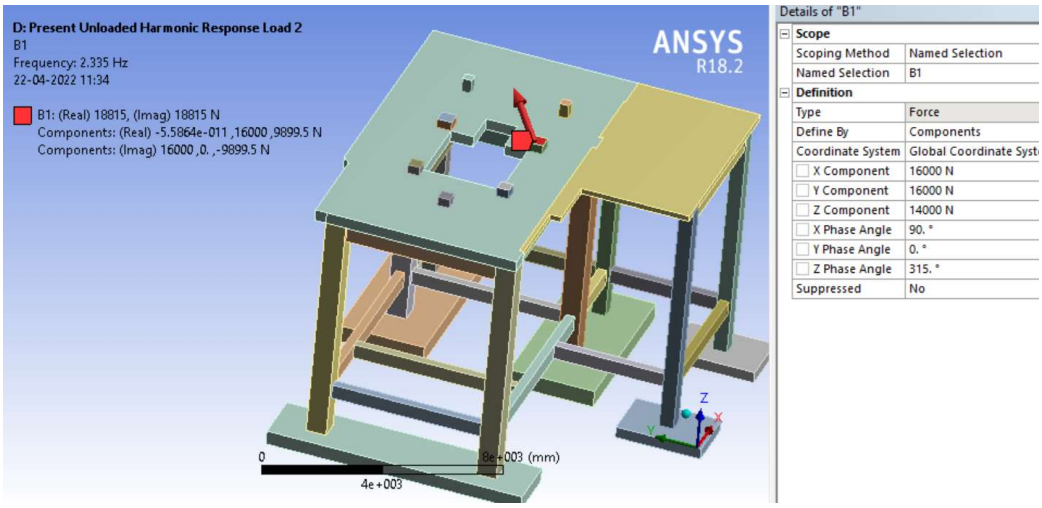


Figure 3g. Excitation Force & Phase angles

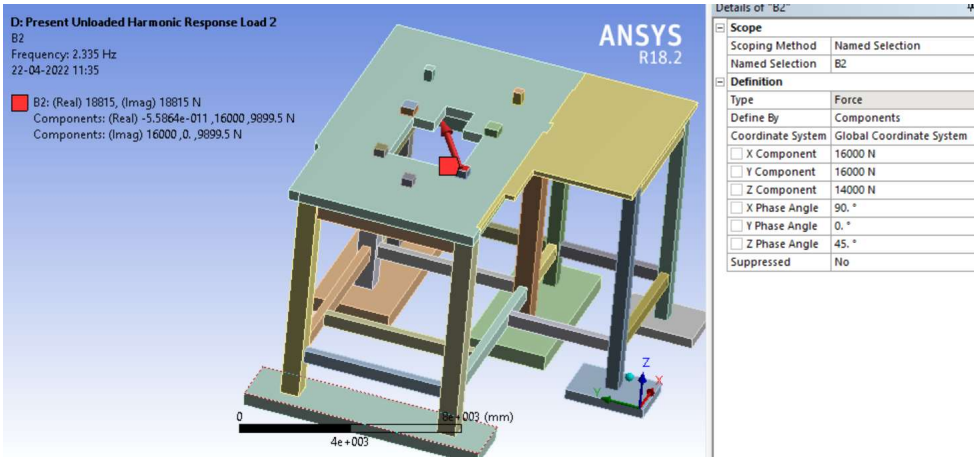


Figure 3h. Excitation Force & Phase angles

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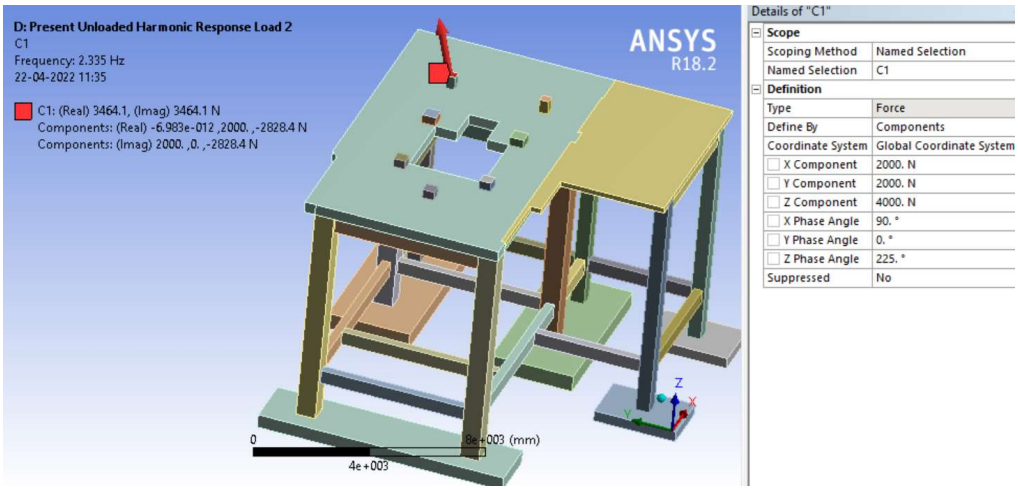


Figure 3i. Excitation Force & Phase angles

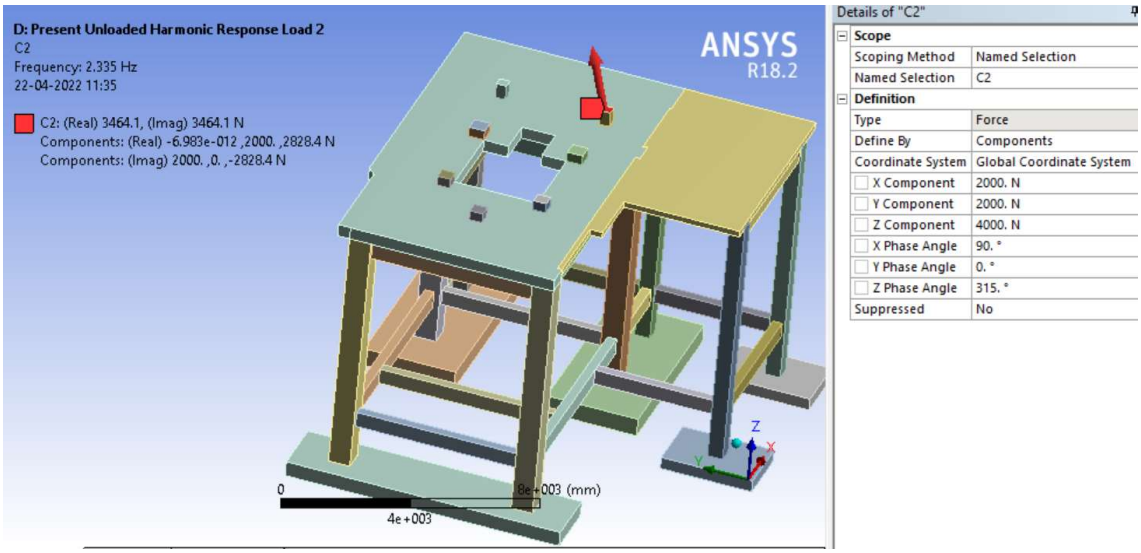


Figure 3j. Excitation Force & Phase angles

3.4. Dynamic Load, Startup & Shutdown

Dynamic analysis is for the startup & shutdown requires time history analysis, which is computationally extensive. Present report is not addressing startup & shutdown conditions. However, if all the natural frequencies are much above the excitation frequencies, startup & shutdown will be very smooth, which can be identified from the FRF of the structure.

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3.5. Boundary Conditions

The model is attached with respective boundary condition at the bottom & sides of the soil, shown in the Figure 4.

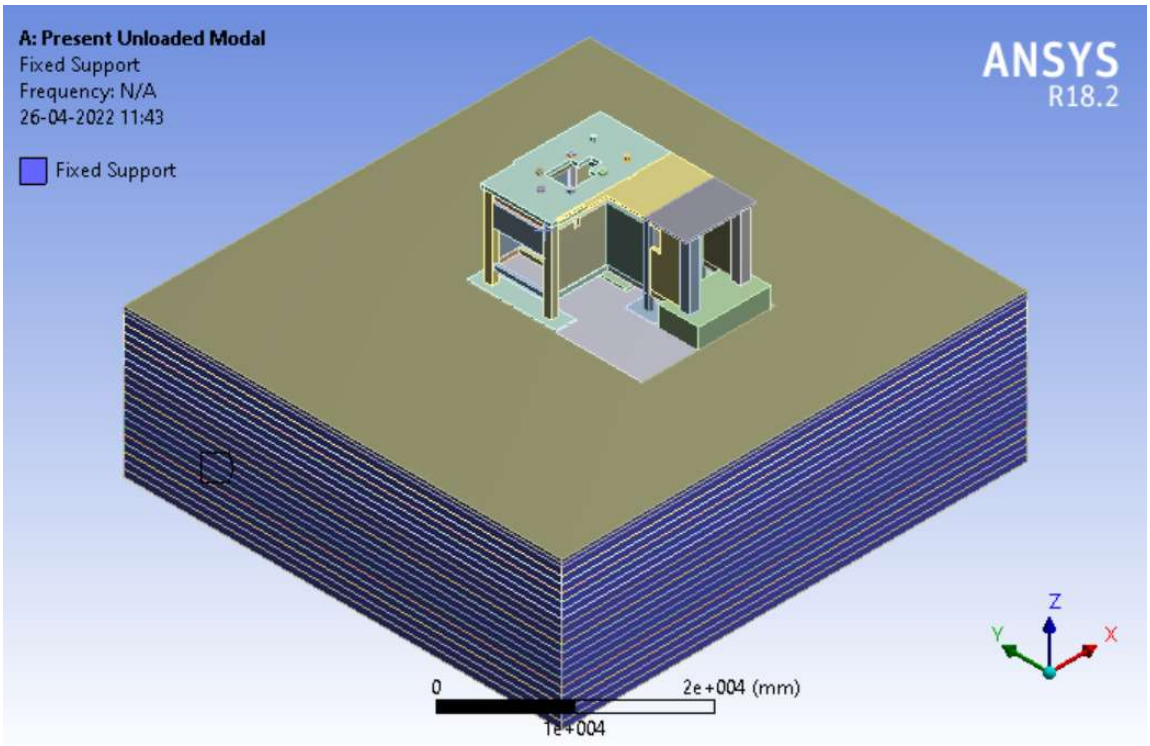


Figure 4. Boundary condition in the soil boundary

4. Analysis of the Existing Structure

Main purpose of the study is to perform harmonic analysis of the existing structure to find the root cause for the excessive structural vibration during operational condition.

4.1. Complete Model

Complete model of the existing system is shown in Figure 5a.

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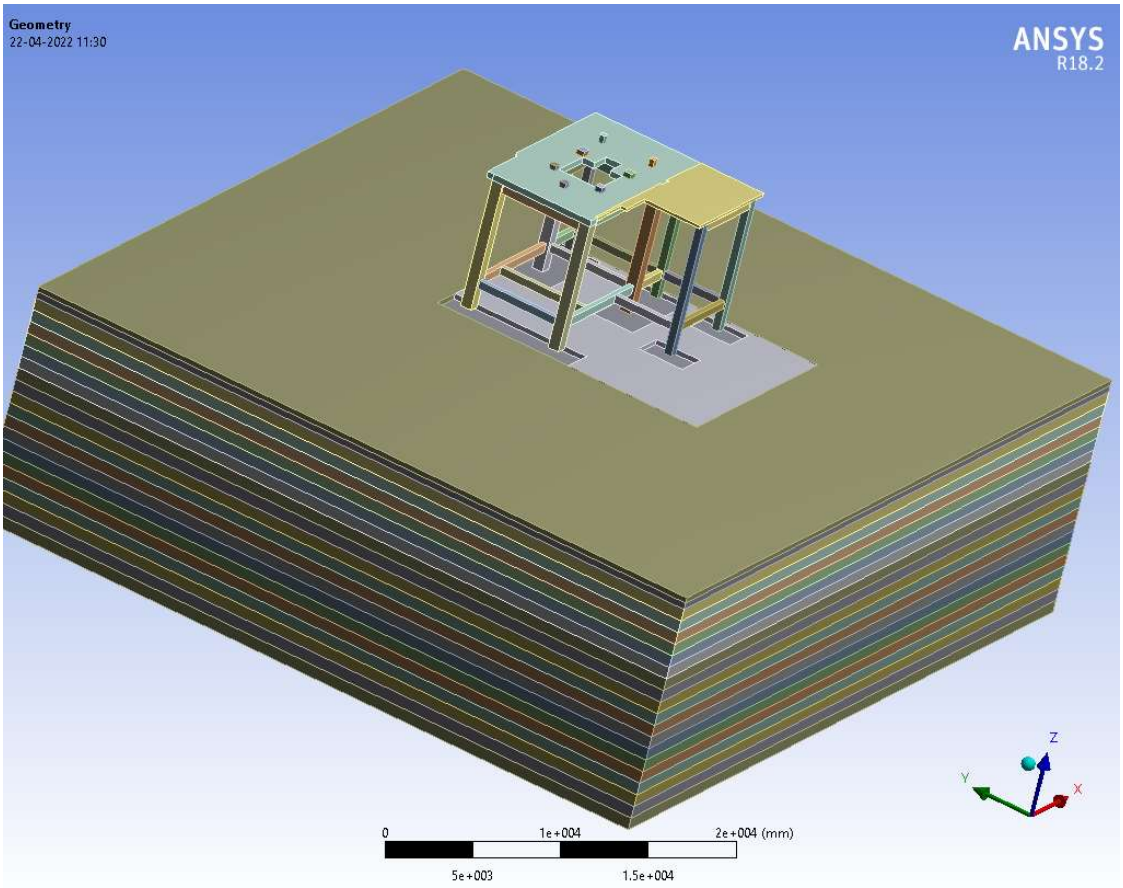


Figure 5a. Complete Model of the Secondary Crusher system along with soil layers

4.2. Mesh

Figures 5b shows the finite element meshes of the Concrete structure & Soil layers.

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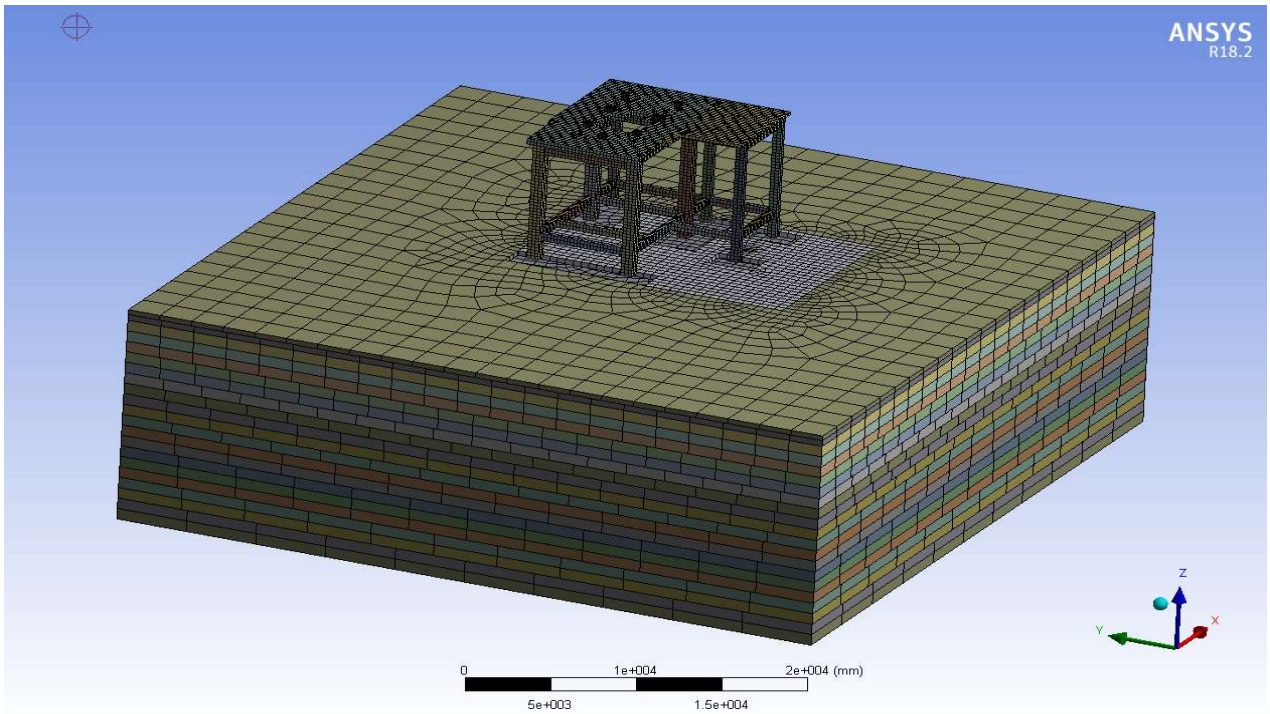
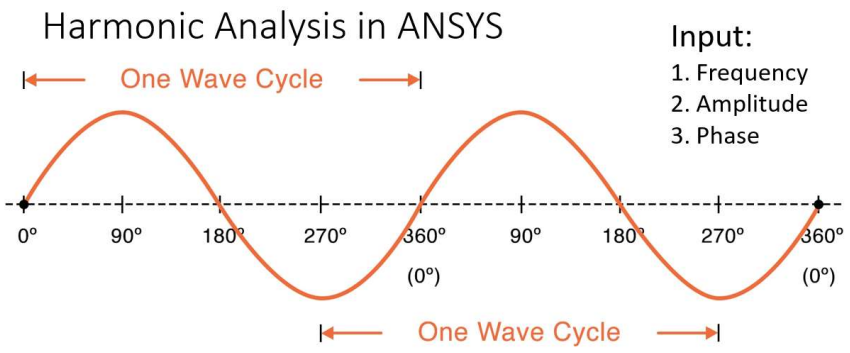


Figure 5b. Finite Element Mesh of Concrete structure

4.3. Harmonic Analysis

Harmonic analysis is performed to determine the steady-state response of the structure subject to the dynamic crusher loads.



Harmonic analyses are **used to determine the steady- state response of a structure to loads that vary sinusoidally (harmonically) with time**.

The main purpose is to investigate if the vibration levels exceeding certain limit when subject to the dynamic load.

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Harmonic analysis of the model provides the response of a system for any sinusoidal loading. Such analyses for a range of frequencies are called frequency response analysis. Plots of different quantities (displacement, velocity, etc.) vs such frequency ranges are called frequency response functions (FRF).

Harmonic analysis is done between up to 9.34 Hz frequencies with 100 intervals, shown in Table 3a.

Details of "Analysis Settings"	
Options	
Frequency Spacing	Linear
Range Minimum	0. Hz
Range Maximum	9.34 Hz
Solution Intervals	100
User Defined Frequencies	Off
Solution Method	Mode Superposition
Include Residual Vector	No
Cluster Results	No
Modal Frequency Range	Program Controlled
Store Results At All Frequencies	Yes

Table 3a. Harmonic analysis settings

4.3.1. Results of Harmonic Analysis

Comparison between measurement & results of harmonic analysis is shown in Table 3b.

Location Mark		Vibration (mm/s)		
Equipment	Compare	Z	Y	X
A2 Pedestal	Measure	1.1	17.8	17.8
	Computed	1.3	22.5	19.5
A1 Pedestal	Measure	0.6	7.3	19.3
	Computed	0.7	16.0	20.0
C1 Pedestal	Measured	1.4	7.7	16.8
	Computed	0.3	11.8	19.5
B2 Pedestal	Measured	1.3	3.5	16.8
	Computed	1.3	22.5	15.5
B1 Pedestal	Measured	2.2	5.5	17.7
	Computed	0.8	16.0	15.4
C2 Pedestal	Measured	1.0	7.7	1.7
	Computed	0.3	11.9	15.5

Table 3b. Harmonic analysis results vs measurement

5. Analysis of the Proposed Structure

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Main purpose of the study is to perform dynamic analysis of the proposed structure to find the structural vibration during operational condition.

5.1. Individual Component of the Model

Crusher is modelled with different components (shown in Figure-6).

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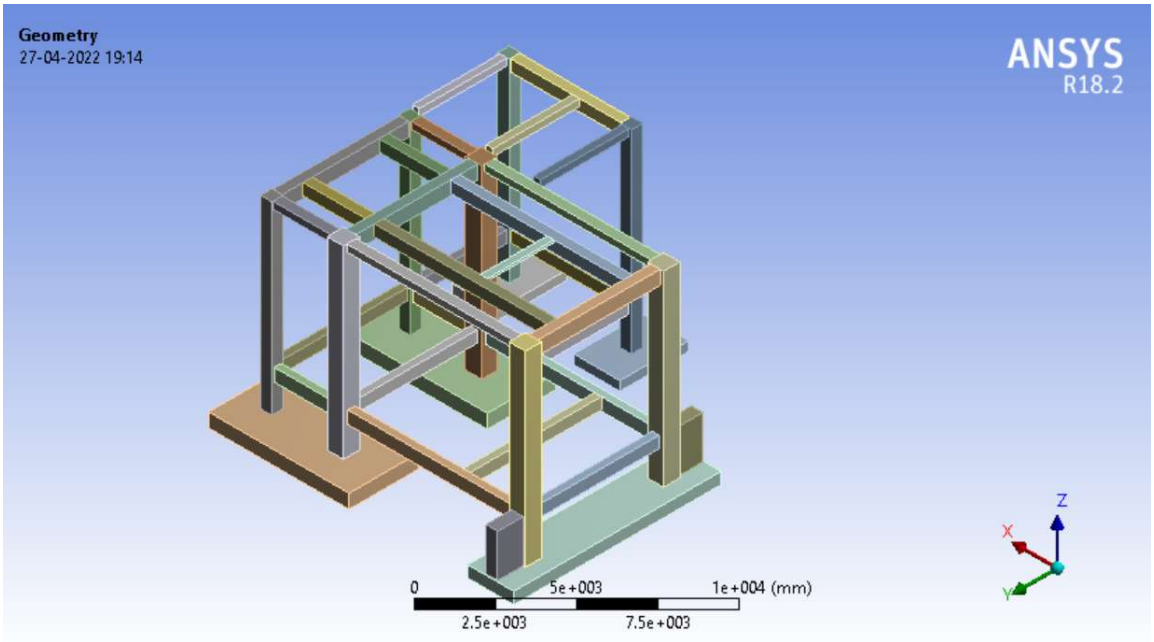
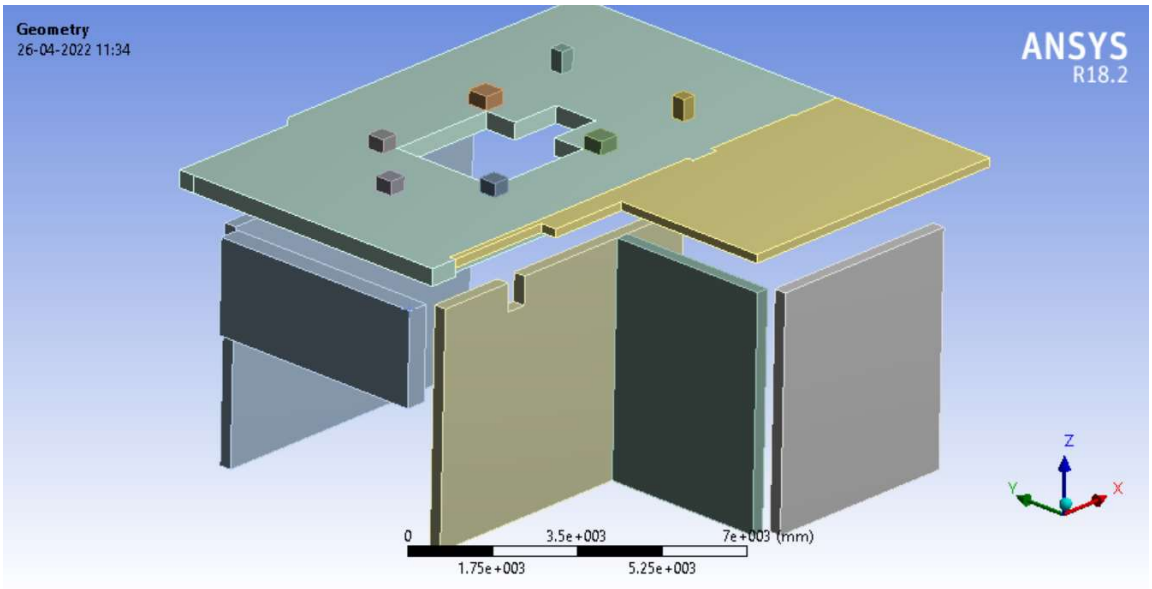


Figure 6a. Footings, beams, columns, & ties of the crusher system.



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Figure 6b. slabs, pedestals, & proposed walls of the crusher system.

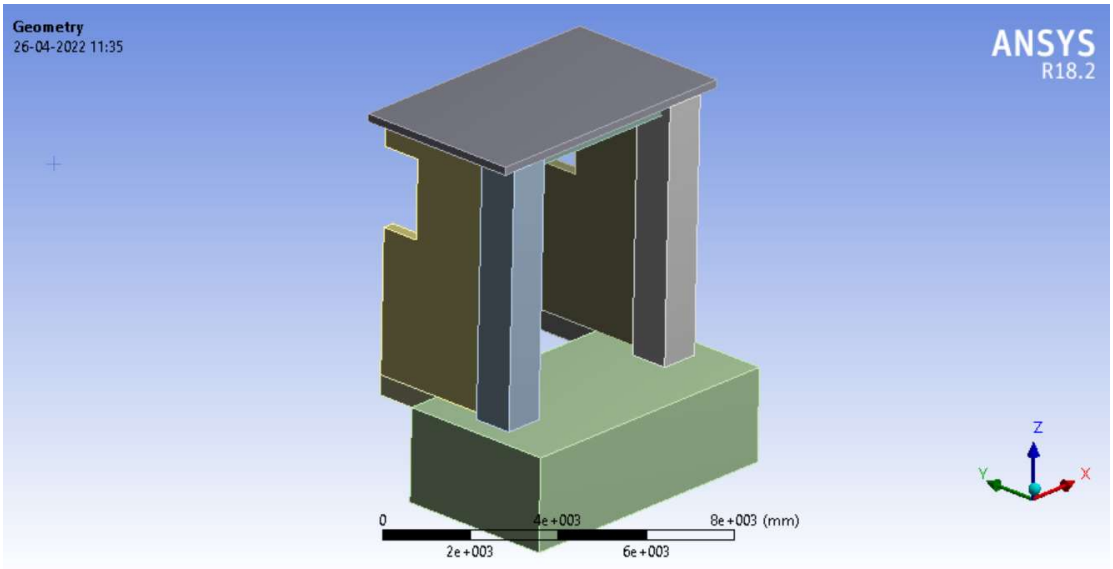


Figure 6c. proposed extension room of the crusher system.

5.2. Complete Model

Complete model of the system is shown in Figure 7.

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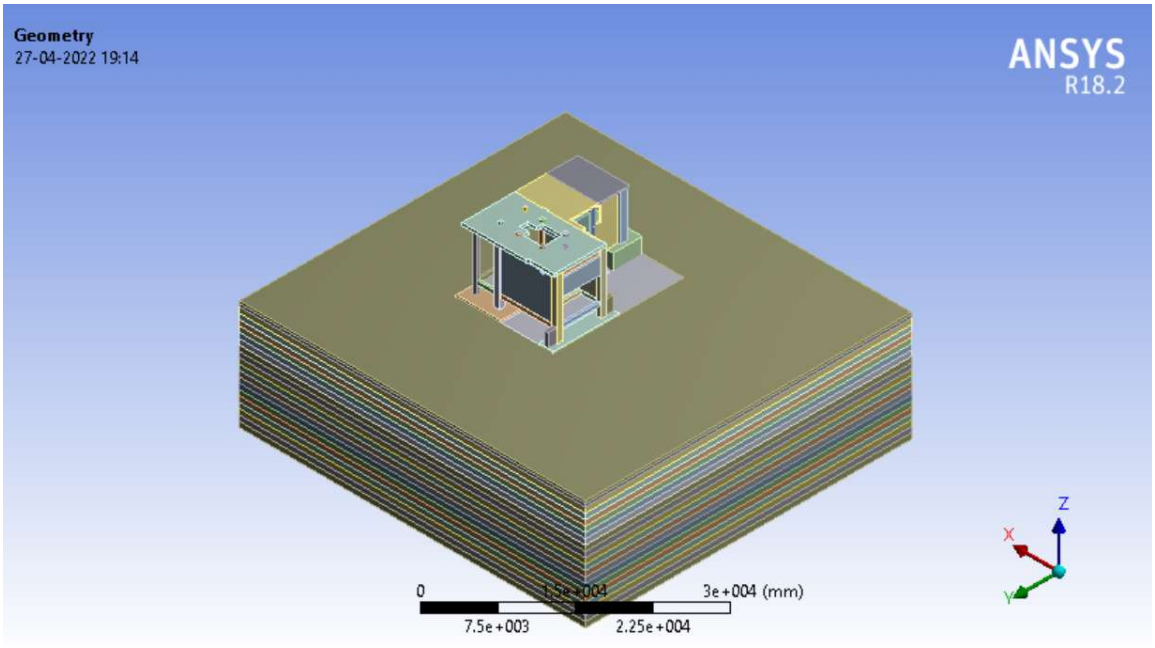


Figure 7a. Complete Model of the Secondary Crusher system along with soil layers

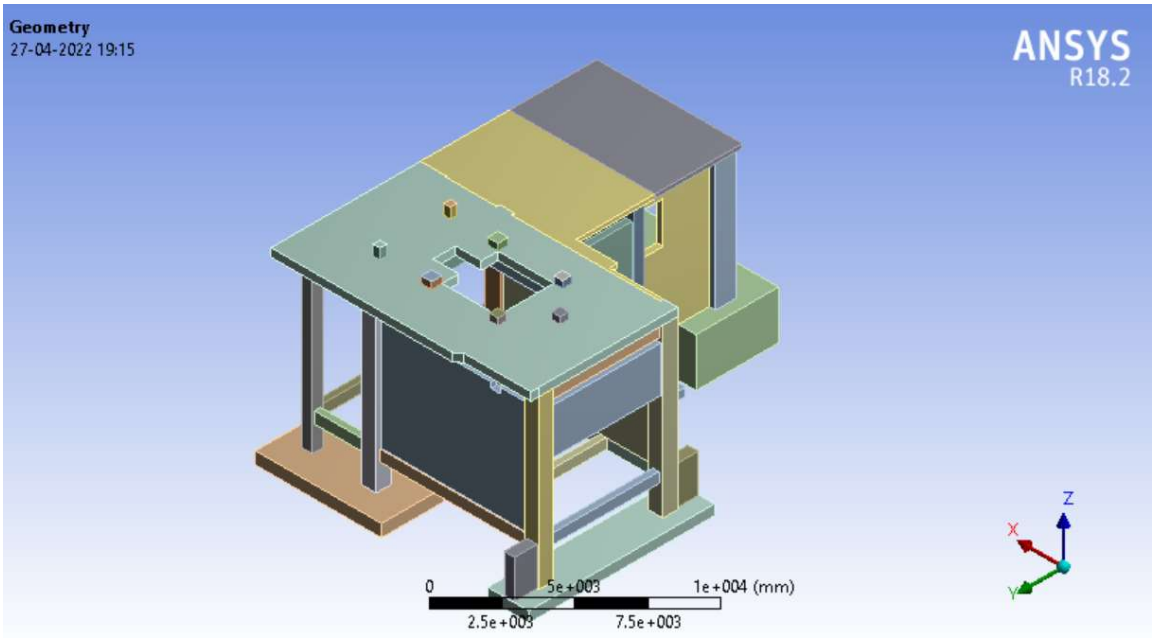


Figure 7b. Complete Model of the Secondary Crusher system without soils

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5.3. Mesh

Figures 8a & 8b shows the finite element meshes of the Concrete structure & Soil layers. The number of elements and nodes are listed in Table 4.

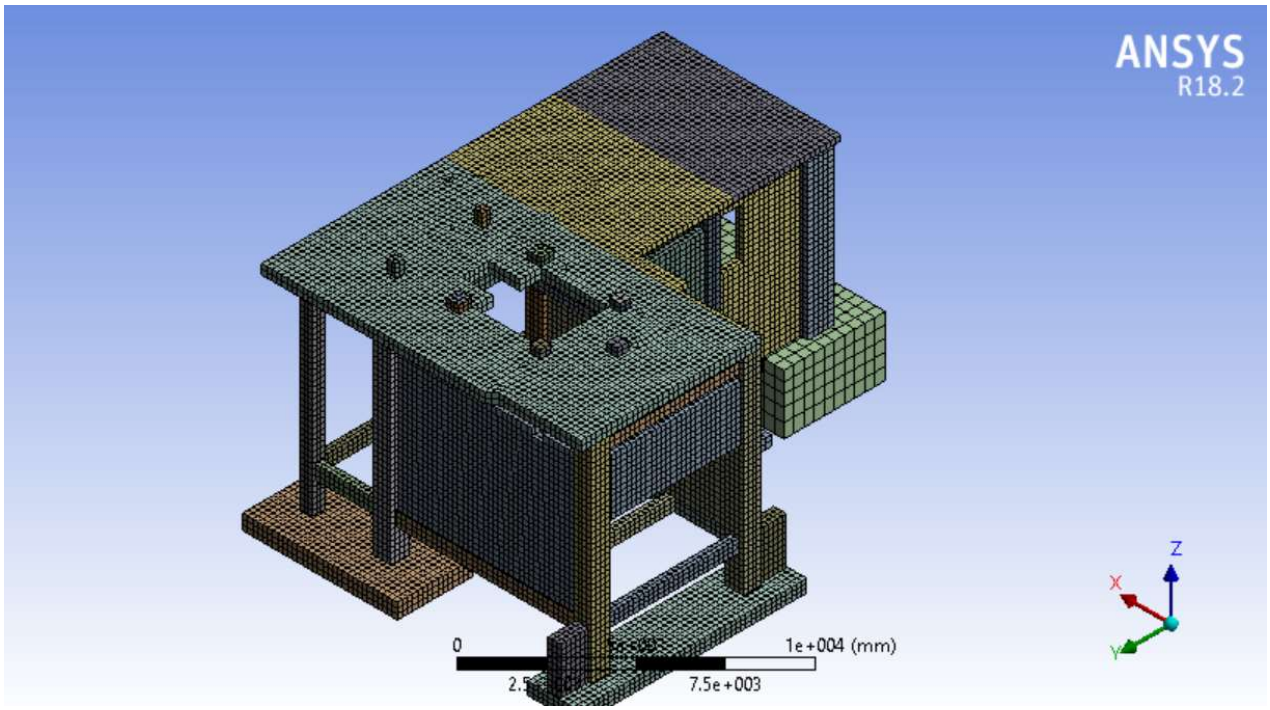


Figure 8a. Finite Element Mesh of Concrete structure

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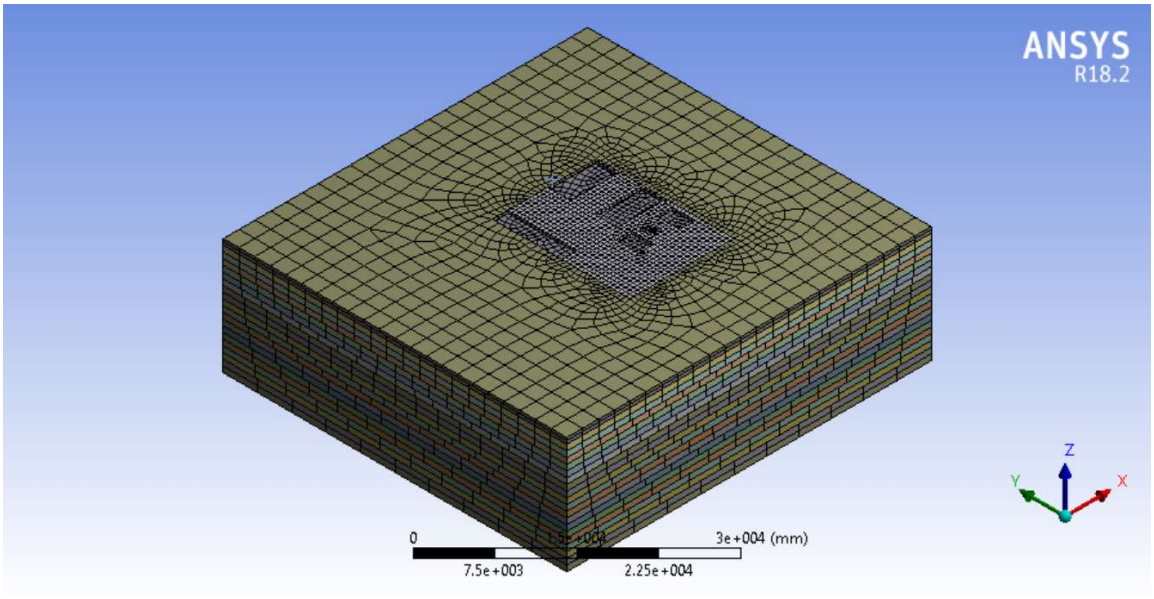


Figure 8b. Finite Element Mesh of soil layers

Statistics	
Nodes	212921
Elements	43122

Table 4. Finite Element Mesh Statistics

5.4. Modal Analysis

Table 5 and Figure-9 shows first 3 natural frequencies and first mode shapes for unloaded & loaded conditions. To verify the safe condition of the system at these frequencies, harmonic response analyses are to be performed.

Secondary Crusher System				
	Unloaded Condition		Loaded Condition	
Mode	Hz	Ratio	Hz	Ratio
1	9.8694	2.11	9.5114	2.04
2	10.829	2.32	10.602	2.27
3	11.564	2.48	11.265	2.41

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Table 5. First 12 Natural Frequencies of Secondary Crusher with Soil layers.

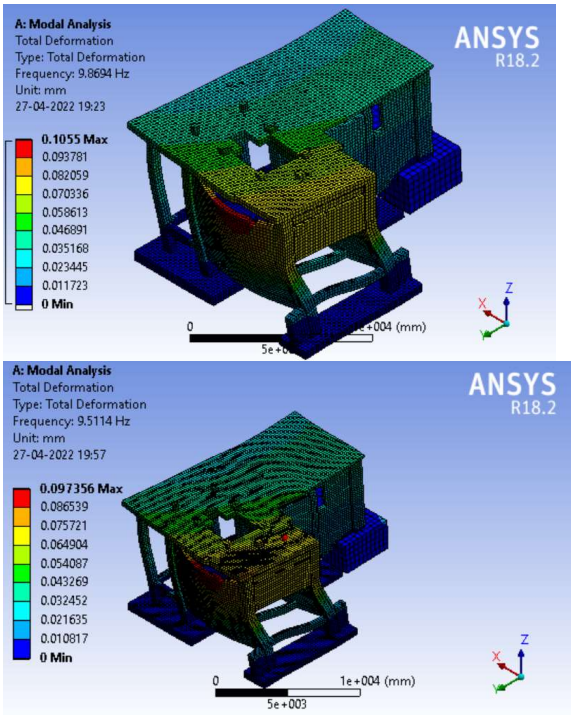


Figure 9. First Mode Shapes of Secondary Crusher: Left: Unloaded. Right: Loaded

5.5. Harmonic Analysis

5.5.1. Relationships between different terms

Relationship between peak velocity vs peak displacement vs peak accelerations is:

Peak velocity = 2 * PI * frequency * peak displacement ……. Equation (1)

Peak acceleration = 2 * PI * frequency * peak velocity ……. Equation (2)

Where for 4.67 Hz frequency, 2 * PI * 4.67 = 29.342 rad/sec

Relationship between peak velocities vs R. M. S. velocities is:

R. M. S. velocity = peak velocity / Sqrt (2) ……. Equation (3)

5.5.2. Phases of the Loadings

Following phase angles are given in different pedestals tops:

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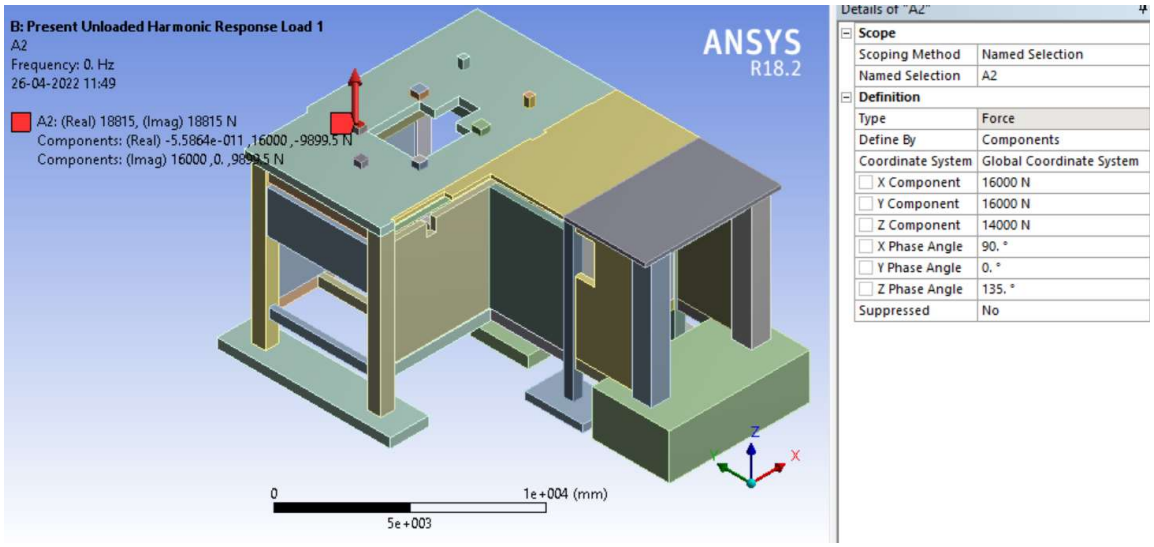


Figure 10a. Excitation Force & Phase angles

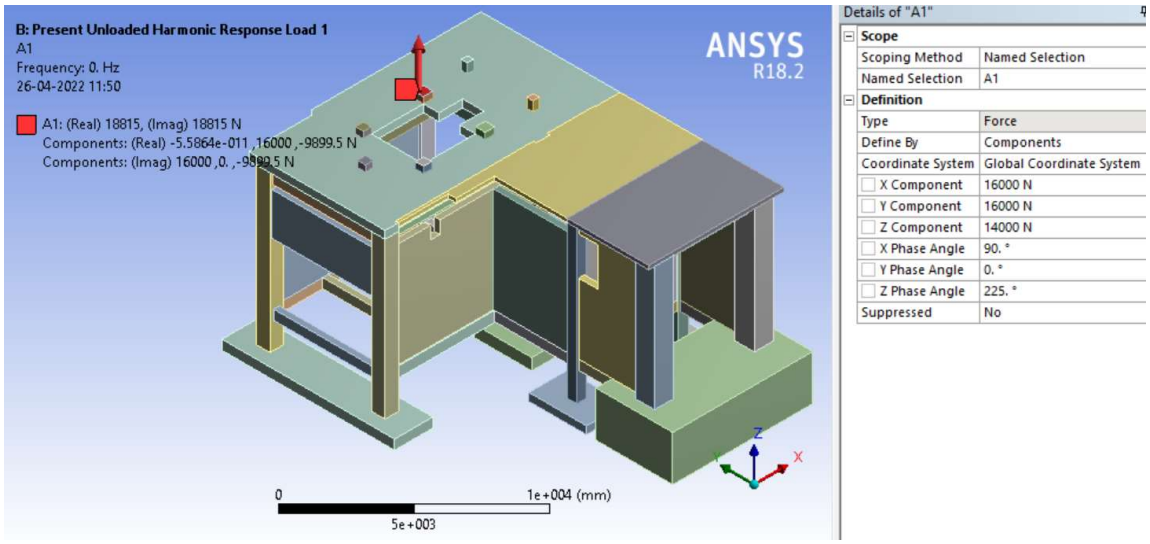


Figure 10b. Excitation Force & Phase angles

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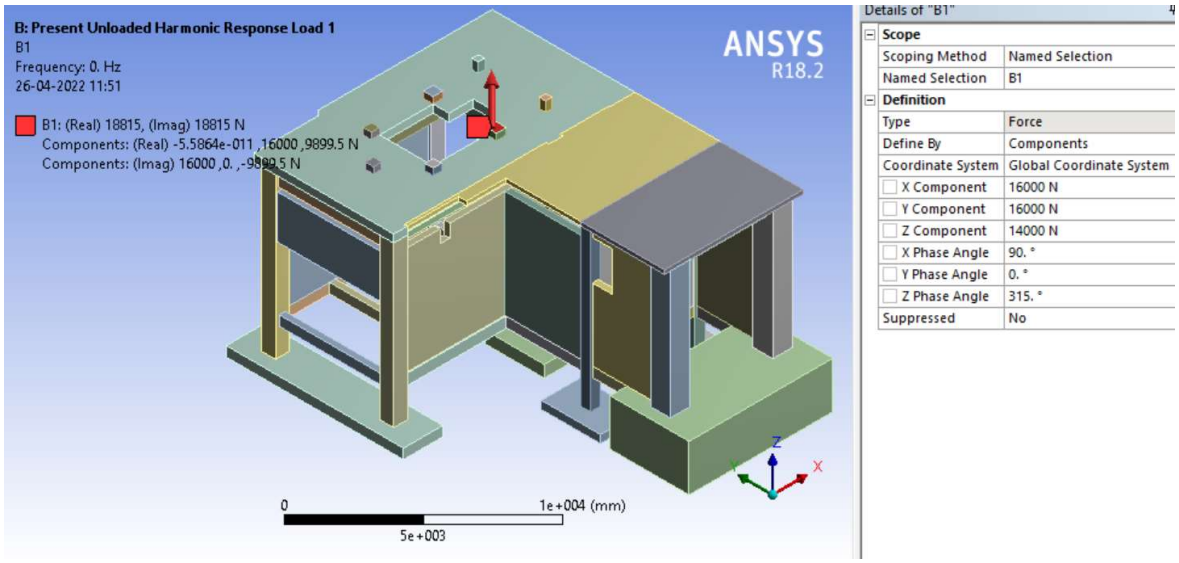


Figure 10c. Excitation Force & Phase angles

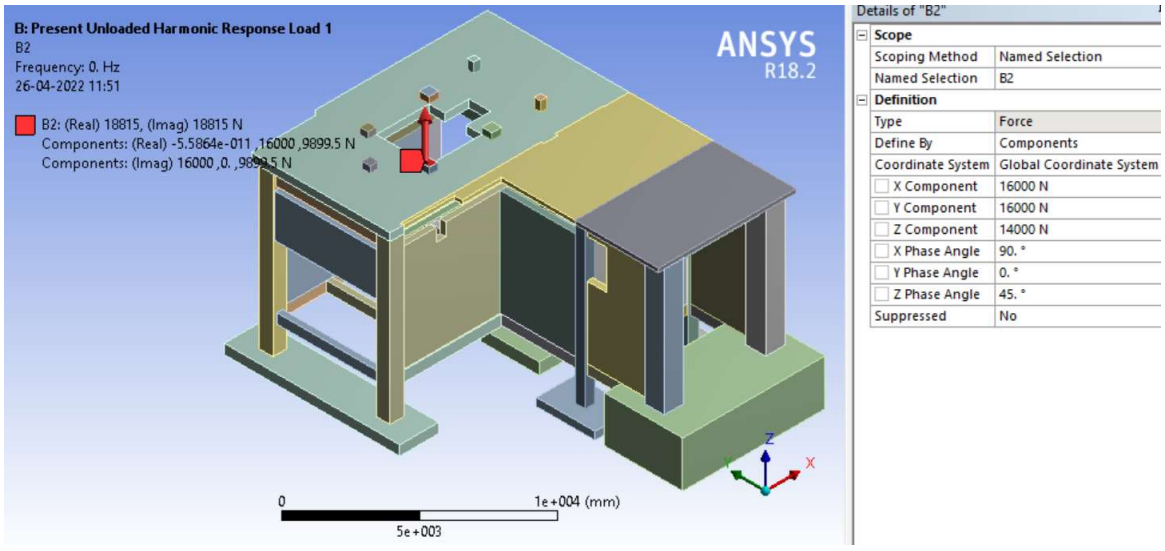


Figure 10d. Excitation Force & Phase angles

Motor is vibrating in horizontal axis. So, following phase angles are carefully chosen to get the synchronal rocking excitation about motor axis.

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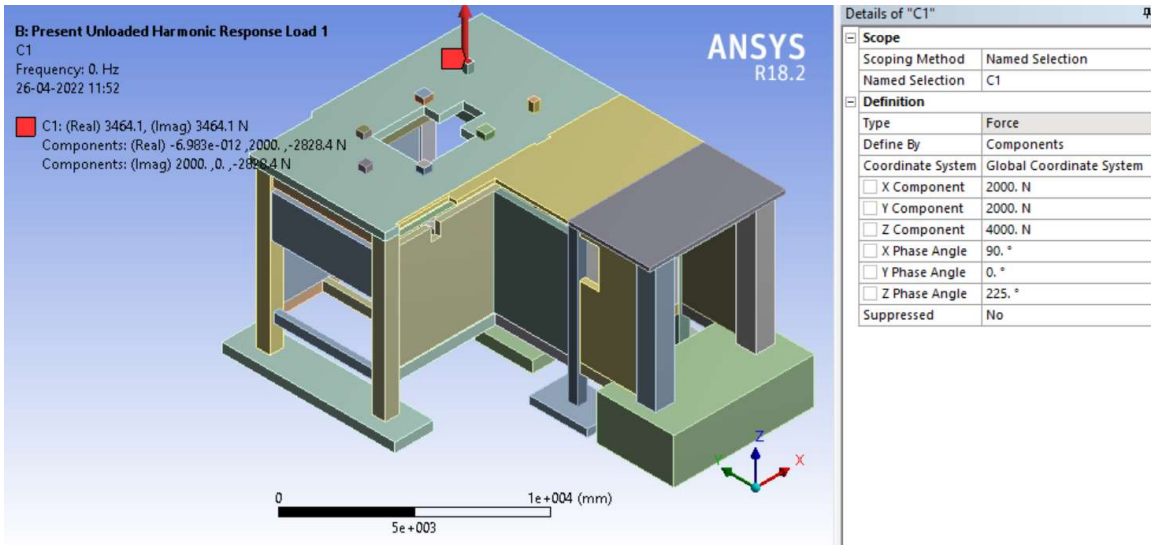


Figure 10e. Excitation Force & Phase angles

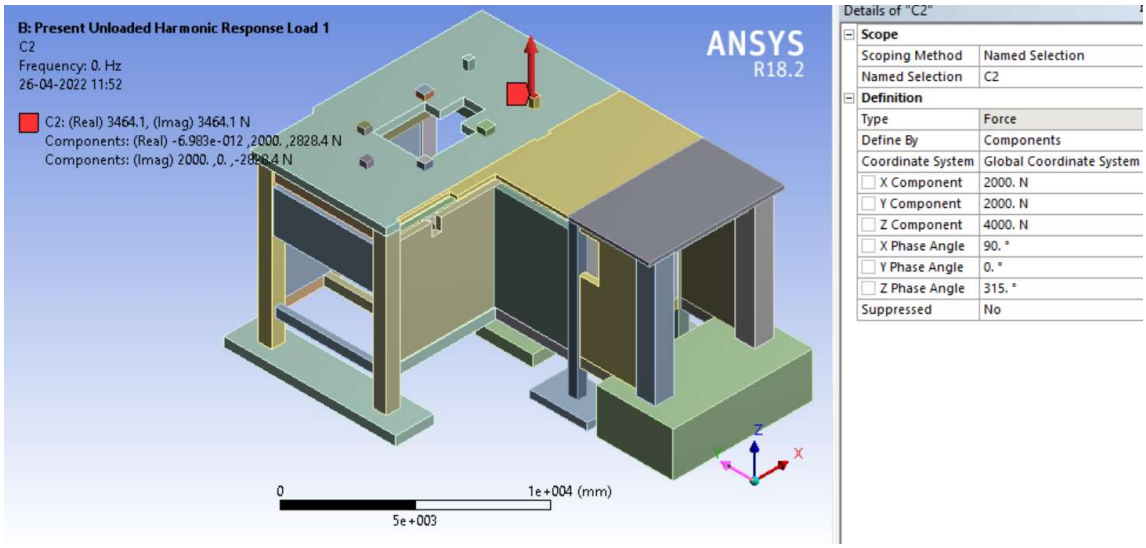


Figure 10f. Excitation Force & Phase angles

5.5.3. Peak Total Velocity of Concrete Structure, Unloaded Condition: (4.67 Hz)

Figure 11 shows the maximum peak total velocities among different phases of the Loadings

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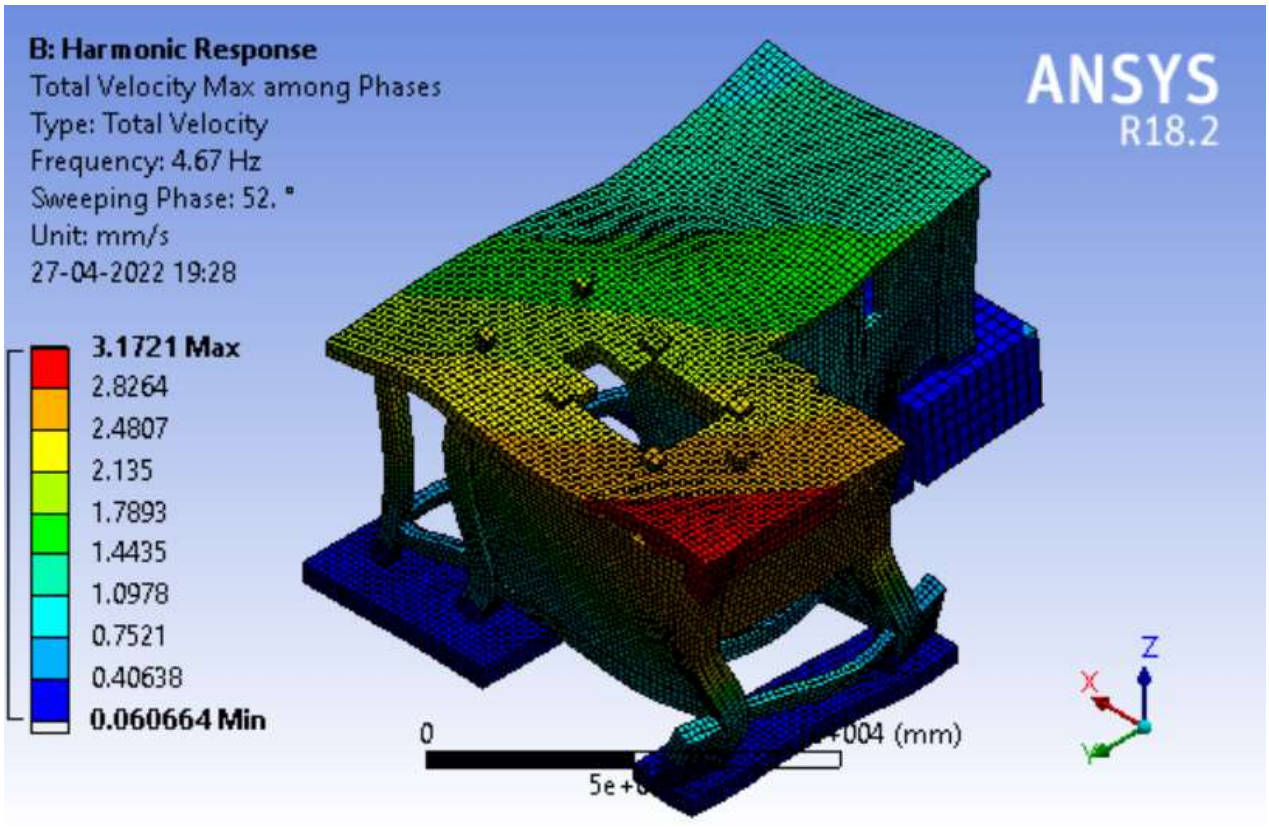
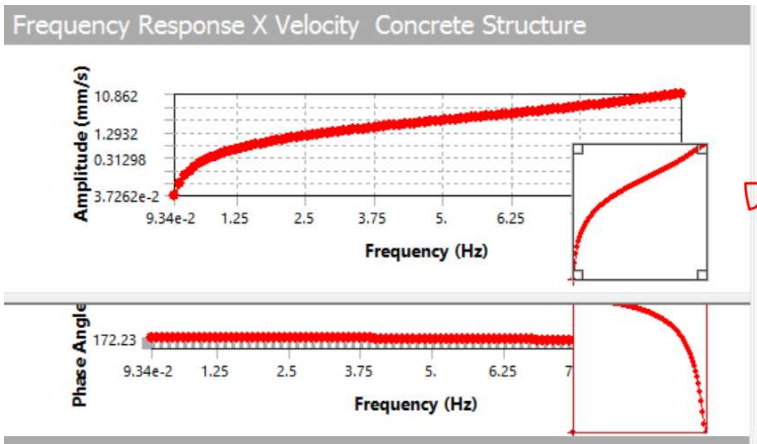


Figure 11. maximum peak total velocity

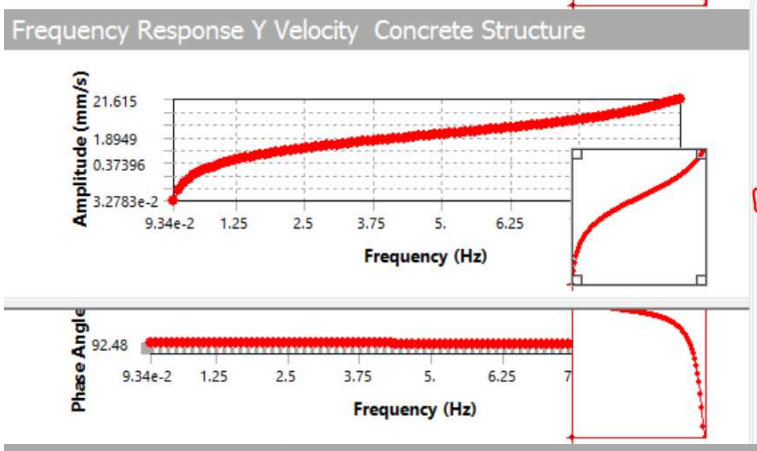
5.5.4. FRF for Peak Velocity of Concrete Structure, Unloaded Condition: (0-9.34 Hz)

Figure 12 shows the FRF of peak directional velocities for different frequencies

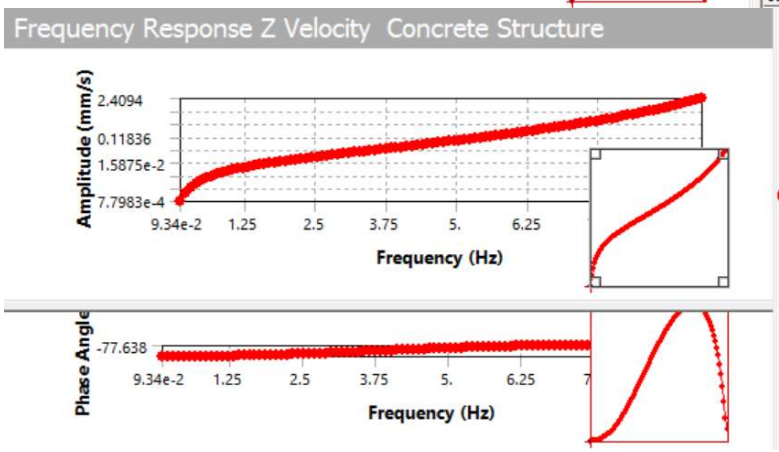
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	Frequency [Hz]	Amplitude [mm/s]	Phase Angle [°]
43	4.0162	1.8358	169.08
44	4.1096	1.8916	168.96
45	4.203	1.9484	168.83
46	4.2964	2.0064	168.71
47	4.3898	2.0656	168.57
48	4.4832	2.126	168.44
49	4.5766	2.1878	168.3
50	4.67	2.251	168.16
51	4.7634	2.3156	168.01
52	4.8568	2.3817	167.86
53	4.9502	2.4494	167.7
54	5.0436	2.5188	167.53
55	5.137	2.59	167.36
56	5.2304	2.663	167.19
57	5.3238	2.7379	167.01
58	5.4172	2.8149	166.82
59	5.5106	2.894	166.62
60	5.604	2.9754	166.42



	Frequency [Hz]	Amplitude [mm/s]	Phase Angle [°]
43	4.0162	1.6672	90.569
44	4.1096	1.7208	90.52
45	4.203	1.7758	90.471
46	4.2964	1.8321	90.421
47	4.3898	1.8899	90.37
48	4.4832	1.9492	90.318
49	4.5766	2.0101	90.265
50	4.67	2.0728	90.211
51	4.7634	2.1373	90.156
52	4.8568	2.2037	90.099
53	4.9502	2.2721	90.041
54	5.0436	2.3427	89.981
55	5.137	2.4156	89.919
56	5.2304	2.4909	89.856
57	5.3238	2.5688	89.791
58	5.4172	2.6494	89.723
59	5.5106	2.733	89.653
60	5.604	2.8196	89.581



	Frequency [Hz]	Amplitude [mm/s]	Phase Angle [°]
43	4.0162	5.3607e-002	-92.91
44	4.1096	5.6191e-002	-92.185
45	4.203	5.8912e-002	-91.464
46	4.2964	6.178e-002	-90.75
47	4.3898	6.4805e-002	-90.043
48	4.4832	6.7997e-002	-89.344
49	4.5766	7.1369e-002	-88.656
50	4.67	7.4933e-002	-87.978
51	4.7634	7.8702e-002	-87.312
52	4.8568	8.2689e-002	-86.66
53	4.9502	8.6912e-002	-86.022
54	5.0436	9.1386e-002	-85.399
55	5.137	9.6129e-002	-84.793
56	5.2304	0.10116	-84.205
57	5.3238	0.1065	-83.635
58	5.4172	0.11217	-83.084
59	5.5106	0.1182	-82.554
60	5.604	0.12462	-82.045

Figure 12. FRF of peak directional velocities

5.5.5. Peak Total Velocity of Concrete Structure, Loaded Condition: (4.67 Hz)

Figure 13 shows the peak total velocities among different phases of the Loadings

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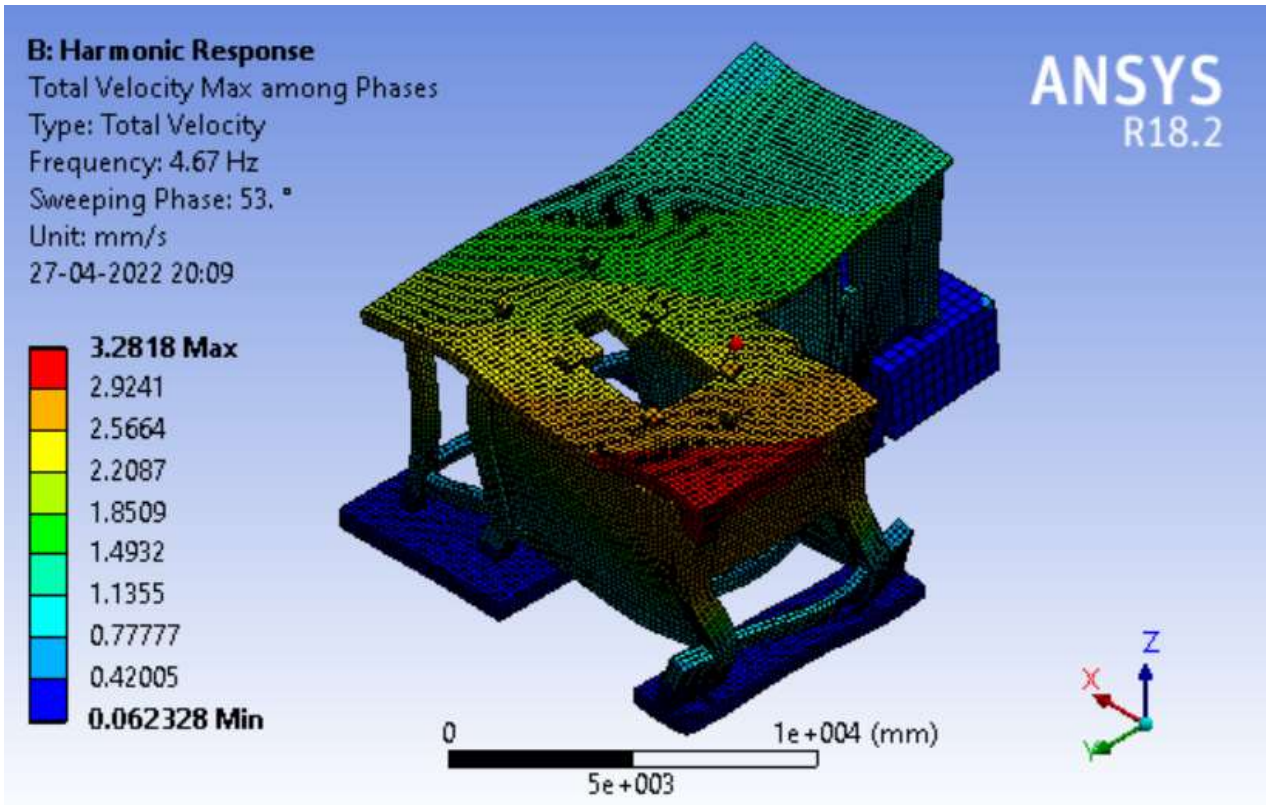
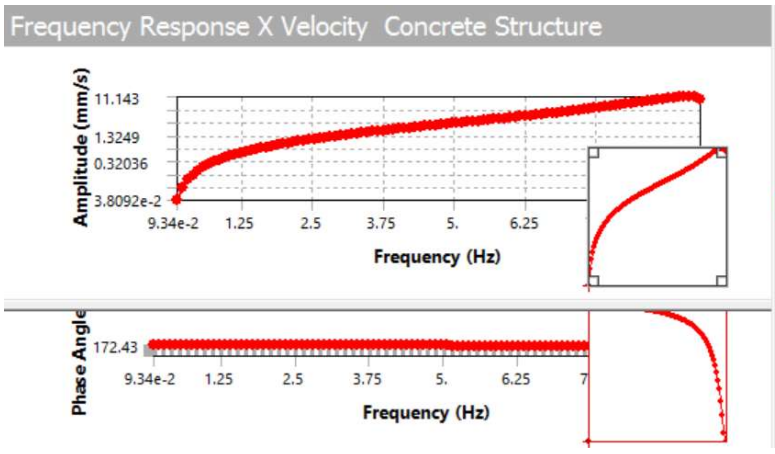


Figure 13. maximum peak total velocity

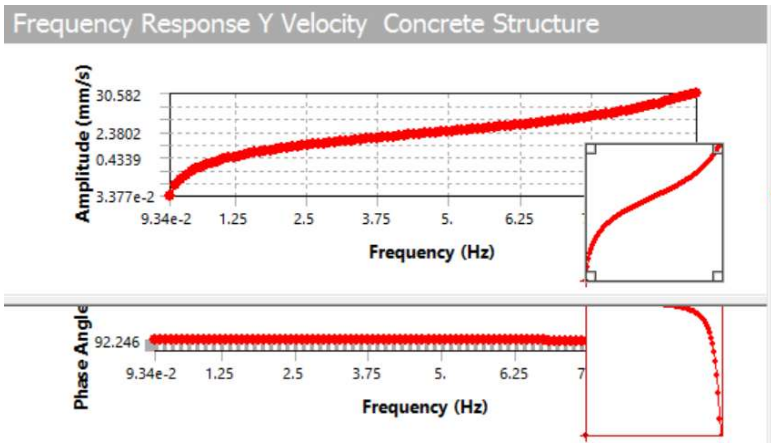
5.5.6. FRF for Peak Directional Velocity of Concrete Structure, Loaded Condition: (0-9.34 Hz)

Figure 12a- 12c shows the FRF of peak directional velocities for different frequencies

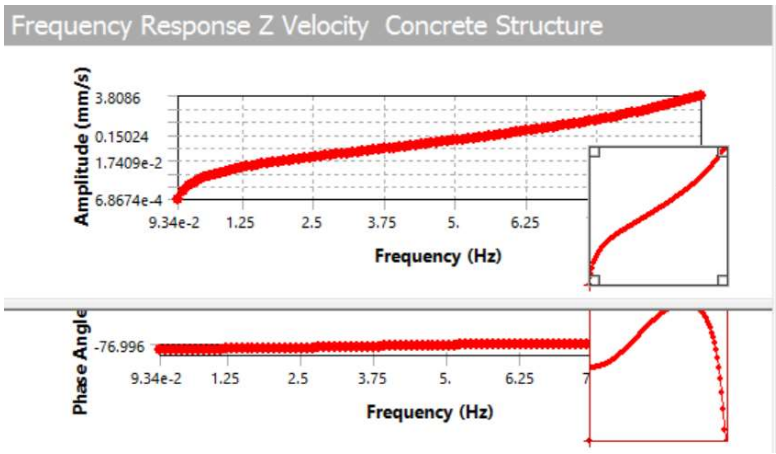


	Frequency [Hz]	Amplitude [mm/s]	Phase Angle [°]
43	4.0162	1.902	169.03
44	4.1096	1.9612	168.9
45	4.203	2.0217	168.76
46	4.2964	2.0835	168.62
47	4.3898	2.1468	168.47
48	4.4832	2.2115	168.32
49	4.5766	2.2778	168.16
50	4.67	2.3457	168.
51	4.7634	2.4153	167.84
52	4.8568	2.4868	167.66
53	4.9502	2.5601	167.48
54	5.0436	2.6355	167.3
55	5.137	2.7129	167.1
56	5.2304	2.7926	166.9
57	5.3238	2.8747	166.7
58	5.4172	2.9592	166.48
59	5.5106	3.0464	166.25
60	5.604	3.1363	166.02

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	Frequency [Hz]	Amplitude [mm/s]	Phase Angle [°]
43	4.0162	1.7421	90.294
44	4.1096	1.7996	90.244
45	4.203	1.8587	90.194
46	4.2964	1.9194	90.142
47	4.3898	1.9817	90.09
48	4.4832	2.0459	90.036
49	4.5766	2.112	89.981
50	4.67	2.1802	89.925
51	4.7634	2.2505	89.867
52	4.8568	2.3231	89.807
53	4.9502	2.3981	89.746
54	5.0436	2.4758	89.683
55	5.137	2.5562	89.618
56	5.2304	2.6395	89.55
57	5.3238	2.726	89.48
58	5.4172	2.8159	89.407
59	5.5106	2.9093	89.332
60	5.604	3.0067	89.252



	Frequency [Hz]	Amplitude [mm/s]	Phase Angle [°]
43	4.0162	5.354e-002	-87.26
44	4.1096	5.6386e-002	-86.652
45	4.203	5.9397e-002	-86.054
46	4.2964	6.2584e-002	-85.466
47	4.3898	6.596e-002	-84.89
48	4.4832	6.9539e-002	-84.326
49	4.5766	7.3334e-002	-83.777
50	4.67	7.7363e-002	-83.242
51	4.7634	8.1641e-002	-82.722
52	4.8568	8.6187e-002	-82.22
53	4.9502	9.1021e-002	-81.734
54	5.0436	9.6165e-002	-81.267
55	5.137	0.10164	-80.819
56	5.2304	0.10748	-80.391
57	5.3238	0.1137	-79.984
58	5.4172	0.12034	-79.599
59	5.5106	0.12743	-79.236
60	5.604	0.13501	-78.896

Figure 14. FRF of peak directional velocities

5.5.7. Summary of Results

Table 6a shows the maximum Peak displacement, Velocity, and acceleration in the concrete structure for unloaded & loaded conditions. Table 6b shows the R.M.S. velocity of the structure.

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		Total	X	Y	Z
		Peak	Peak	Peak	Peak
Maximum Velocities (mm/sec)	Unloaded	3.172	2.251	2.073	0.075
	Loaded	3.282	2.346	2.180	0.077
Maximum Displacement (micron)	Unloaded	108.106	76.715	70.642	2.553
	Loaded	111.845	79.942	74.302	2.634
Maximum Acceleration (m/sec2)	Unloaded	0.093	0.066	0.061	0.002
	Loaded	0.096	0.069	0.064	0.002

Table 6a. Peak Displacement, Velocity, & Acceleration

		Total	X	Y	Z
		RMS	RMS	RMS	RMS
Maximum Velocities (mm/sec)	Unloaded	**2.243**	**1.592**	**1.466**	**0.053**
	Loaded	**2.321**	**1.659**	**1.542**	**0.055**

Table 6b. RMS Velocity of the structure

6. Conclusions

The maximum peak vibration level for the x, y and z directions over the excitation forcing frequency4.67 Hz is computed in the finite element model. The absolute maximum R.M.S. velocity level computed for frequency is 2.243 & 2.321 mm/s for unloaded & loaded condition respectively. Peak displacements computed 108 & 111 microns for unloaded & loaded conditions respectively.

7. References

[1] ANSYS Mechanical Software

[2] MANSOURAH - MASSARAH GOLD PROJECT GENERAL ARRANGEMENT SECONDARY CRUSHING SECONDARY CONE CRUSHER 3120-CRH-001 (MD - 216 -3120 - EG- ME -GAD - V500-1001) Rev -C01.

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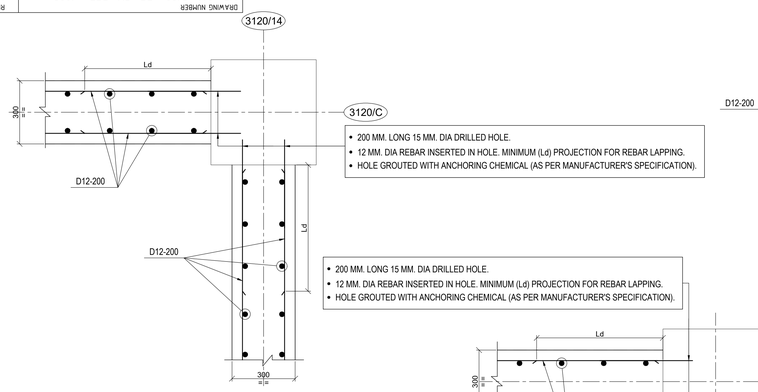
8. Appendix 1

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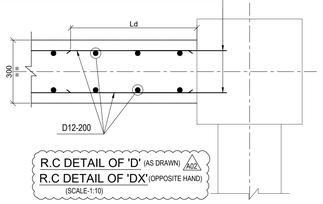
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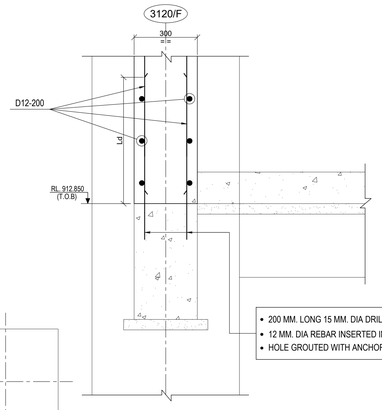
- 200 MM. LONG 15 MM. DIA DRILLED HOLE.
- 12 MM. DIA REBAR INSERTED IN HOLE. MINIMUM (Ld) PROJECTION FOR REBAR LAPPING.
- HOLE GROUTED WITH ANCHORING CHEMICAL (AS PER MANUFACTURER'S SPECIFICATION).

DETAIL-A
(SCALE-1:10)



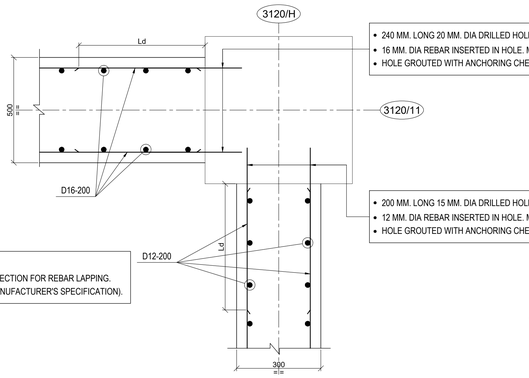
- 200 MM. LONG 15 MM. DIA DRILLED HOLE.
- 12 MM. DIA REBAR INSERTED IN HOLE. MINIMUM (Ld) PROJECTION FOR REBAR LAPPING.
- HOLE GROUTED WITH ANCHORING CHEMICAL (AS PER MANUFACTURER'S SPECIFICATION).

R.C DETAIL OF 'D' (AS DRAWN) AND
R.C DETAIL OF 'DX' (OPPOSITE HAND)
(SCALE-1:10)



- 200 MM. LONG 15 MM. DIA DRILLED HOLE.
- 12 MM. DIA REBAR INSERTED IN HOLE. MINIMUM (Ld) PROJECTION FOR REBAR LAPPING.
- HOLE GROUTED WITH ANCHORING CHEMICAL (AS PER MANUFACTURER'S SPECIFICATION).

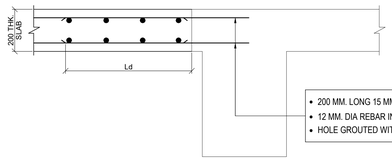
DETAIL-B
(SCALE-1:10)



- 240 MM. LONG 20 MM. DIA DRILLED HOLE.
- 16 MM. DIA REBAR INSERTED IN HOLE. MINIMUM (Ld) PROJECTION FOR REBAR LAPPING.
- HOLE GROUTED WITH ANCHORING CHEMICAL (AS PER MANUFACTURER'S SPECIFICATION).

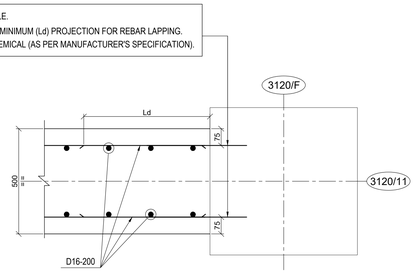
- 200 MM. LONG 15 MM. DIA DRILLED HOLE.
- 12 MM. DIA REBAR INSERTED IN HOLE. MINIMUM (Ld) PROJECTION FOR REBAR LAPPING.
- HOLE GROUTED WITH ANCHORING CHEMICAL (AS PER MANUFACTURER'S SPECIFICATION).

SECTION Z3-Z3
(SCALE-1:10)

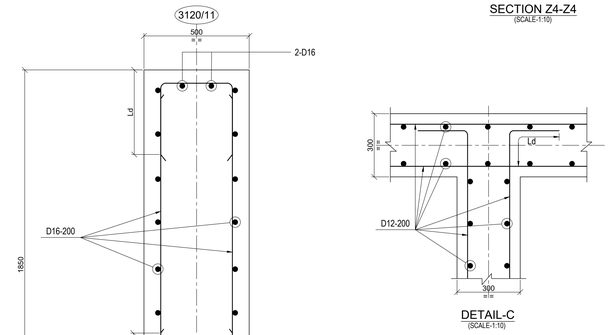


- 200 MM. LONG 15 MM. DIA DRILLED HOLE.
- 12 MM. DIA REBAR INSERTED IN HOLE. MINIMUM (Ld) PROJECTION FOR REBAR LAPPING.
- HOLE GROUTED WITH ANCHORING CHEMICAL (AS PER MANUFACTURER'S SPECIFICATION).

SECTION Z-Z
(R.C.C DETAILS)
(SCALE-1:10)

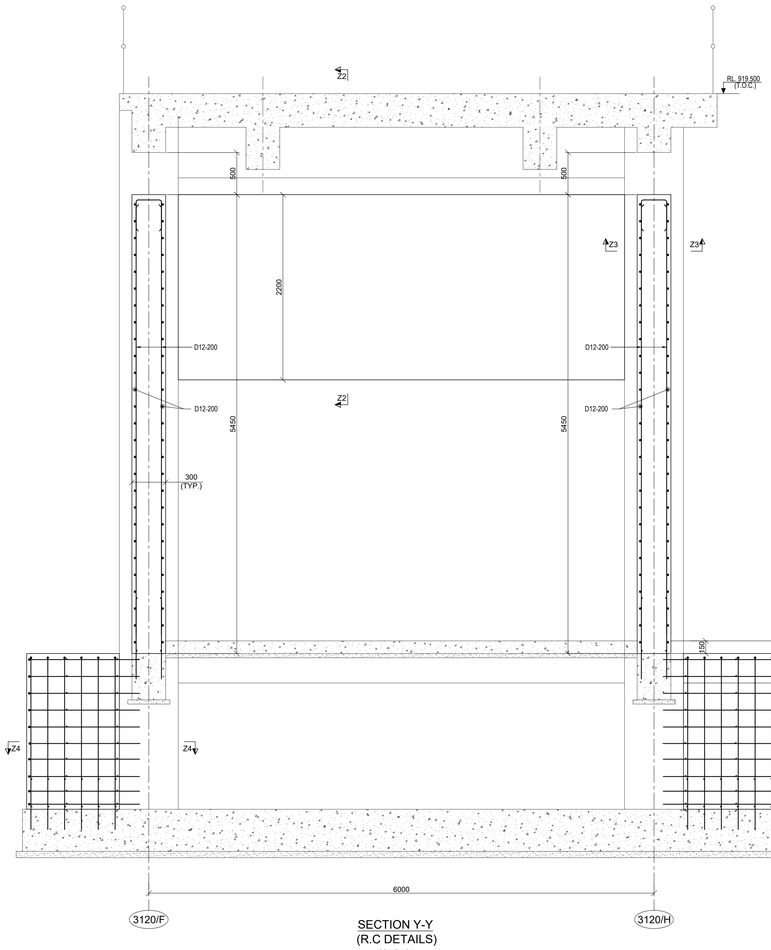


SECTION Z4-Z4
(SCALE-1:10)

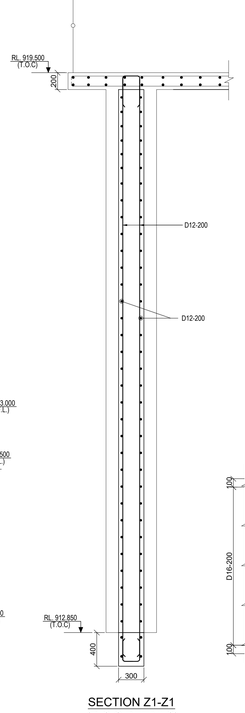


- 240 MM. LONG 20 MM. DIA DRILLED HOLE.
- 16 MM. DIA REBAR INSERTED IN HOLE. MINIMUM (Ld) PROJECTION FOR REBAR LAPPING.
- HOLE GROUTED WITH ANCHORING CHEMICAL (AS PER MANUFACTURER'S SPECIFICATION).

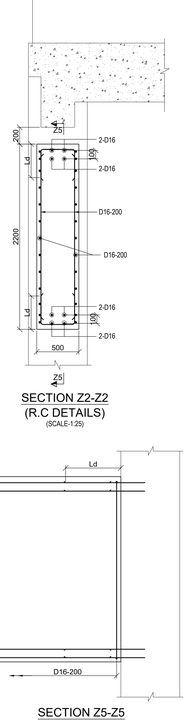
DETAIL-C
(SCALE-1:10)



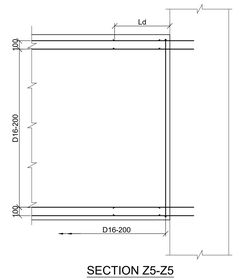
SECTION Y-Y
(R.C DETAILS)
(SCALE-1:25)



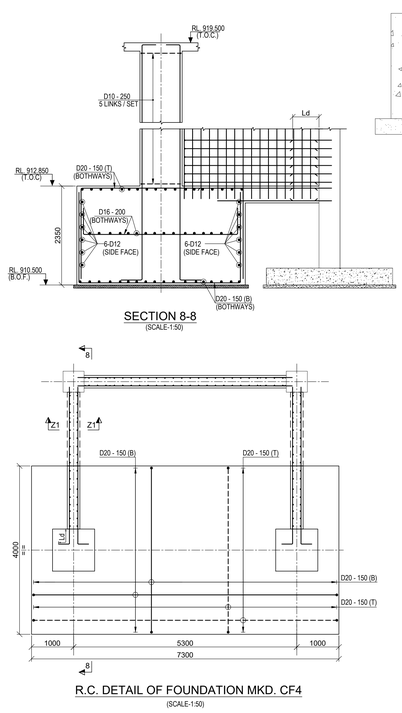
SECTION Z1-Z1
(R.C.C DETAILS)
(SCALE-1:25)



SECTION Z2-Z2
(R.C DETAILS)
(SCALE-1:25)

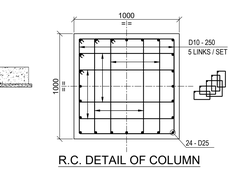


SECTION Z5-Z5
(R.C DETAILS)
(SCALE-1:25)

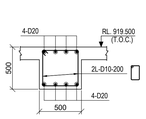


SECTION 8-8
(SCALE-1:50)

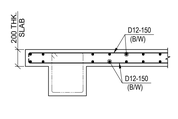
R.C. DETAIL OF FOUNDATION MKD. CF4
(SCALE-1:50)



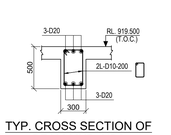
R.C. DETAIL OF COLUMN
MKD. C3
SCALE 1:20



TYP. CROSS SECTION OF
BEAM MKD. NB1
(SLAB REIN. NOT SHOWN FOR CLARITY)
(SCALE- 1:25)



TYP. R.C. DETAIL OF ROOF SLAB
(BEAM REIN. NOT SHOWN FOR CLARITY)
(SCALE- 1:25)



TYP. CROSS SECTION OF
BEAM MKD. NB2
(SLAB REIN. NOT SHOWN FOR CLARITY)
(SCALE- 1:25)

LEGENDS :-
- RL - REDUCED LEVEL
- T.O.C. - TOP OF CONCRETE
- F.G.L - FINISHED GROUND LEVEL
- F.F.L - FINISHED FLOOR LEVEL
- B.O.F - BOTTOM OF FOUNDATION
- TYP. - TYPICAL
- R.C.C - REINFORCED CEMENT CONCRETE
- P.C.C - PLAIN CEMENT CONCRETE
- THK. - THICK
- D - DIAMETER OF BAR
- B/W - BOTH WAYS
- BOTTOM REINFORCEMENT (B) : ———
- TOP REINFORCEMENT (T) : – – – – –

NOTES :-
1. ALL DIMENSIONS ARE IN MILLIMETRES AND LEVELS ARE IN METRES.
2. GRADE OF CONCRETE SHALL BE C35.
4. GRADE OF REINFORCEMENT SHALL BE DEFORMED BAR ASTM A615 - GRADE 60 (fy = 420 MPa).
5. NET SAFE BEARING CAPACITY OF SOIL AT BOTTOM OF FOUNDATION CONSIDERED 500KN/SQ. M.
6. CLEAR COVER TO MAIN REINFORCEMENT SHALL BE AS
   a) FOUNDATION = 75 mm (BOTTOM & SIDE) 50 mm. (TOP)
   b) COLUMN & PEDESTAL = 50mm
   c) BEAM = 50mm
   d) SLAB = 40mm
   e) WALL = 40 mm ABOVE F.G.L & 75 mm BELOW F.G.L
7. MINIMUM LAP LENGTH OF REINFORCEMENT SHALL BE 40 X DIA. OF BAR.
8. QUICK SETTING ADMIXTURE SHALL BE USED FOR REDUCING THE SETTING TIME OF THE CONCRETE.
9. DO NOT SCALE. PLEASE ASK.

REV	DESCRIPTION	DATE	DRN	CHK	ENG	APP	EMP
A02	REVISED AS PER EMPLOYER COMMENTS DATED 06/06/2022	14/10/2022	SYA	SBY	APR	AG	
A01	ISSUED OF APPROVAL	28/04/2022	SYA	SBY	APR	AG	

MD-216-3120-EG-CV-GAD-1011 — MODIFICATION OF CRUSHER SUPPORT STRUCTURE

DOCUMENT NUMBER | REFERENCE DOCUMENTS

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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LARSEN & TOUBRO DRAWING NUMBER: 019045_100_C_G_24770_001 — REVISION: 1
OUTOTEC DRAWING NUMBER: 900707__GAU01__ZC01__CTB17__31234 — DOCUMENT ID: OU100751400 — REVISION: 1

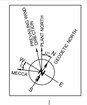
DRAWN	SYA	18/04/2022
CHECKED	SBY	20/04/2022
ENGINEER	APR	25/04/2022
APPROVED	AG	28/04/2022
EMPLOYER		

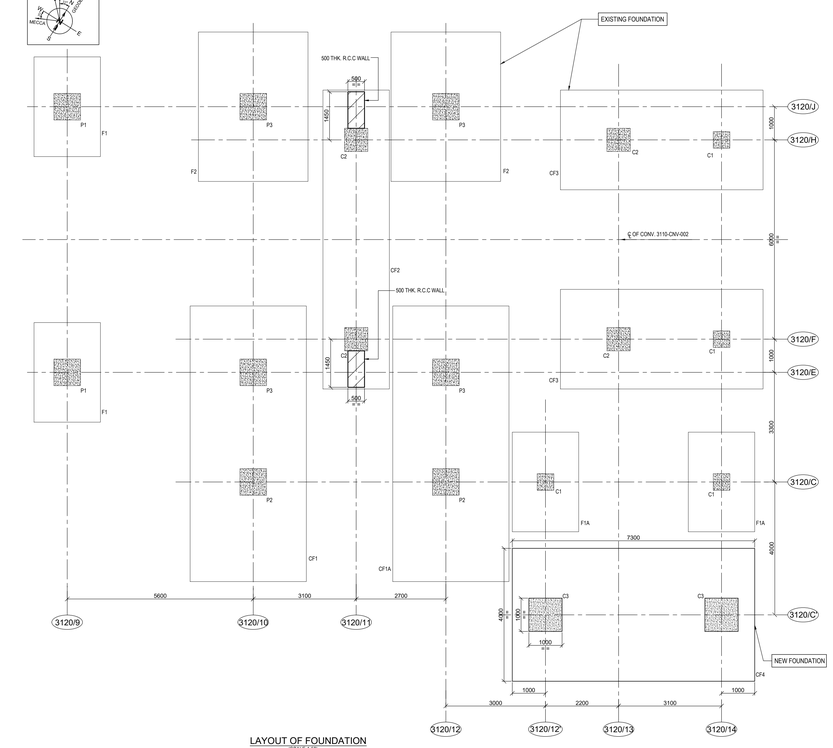
SCALE: 1:50,1:25,1:10 — A0

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
SECONDARY CRUSHING
R.C DETAILS OF ADDITIONAL FOUNDATION & WALL

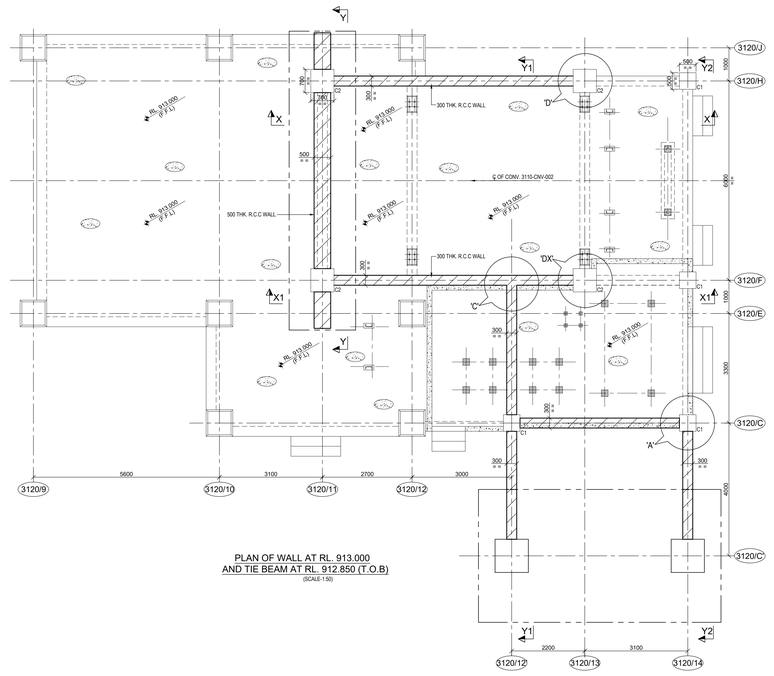
DRAWING NUMBER: MD-216-3120-EG-CV-DED-1009 — REV: A02

DRAWING NUMBER: MD-216-3120-EG-CV-GAD-1011 | REV: A03

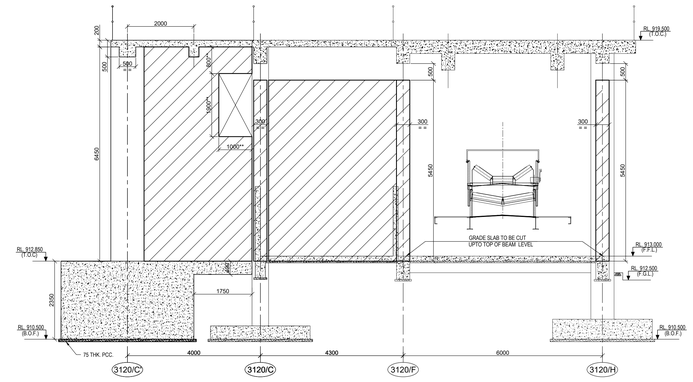




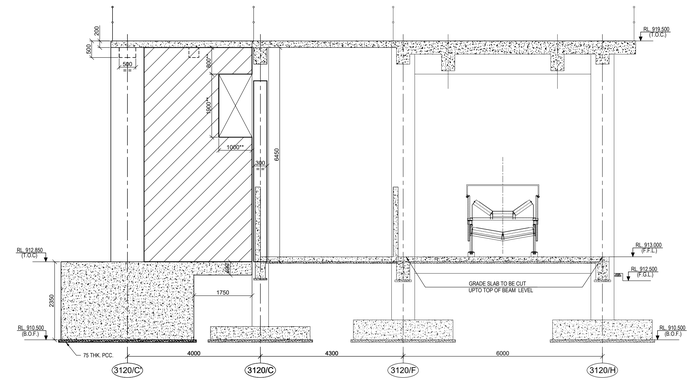
LAYOUT OF FOUNDATION
(SCALE-1:50)



PLAN OF WALL AT RL. 913.000
AND TIE BEAM AT RL. 912.850 (T.O.B)
(SCALE-1:50)

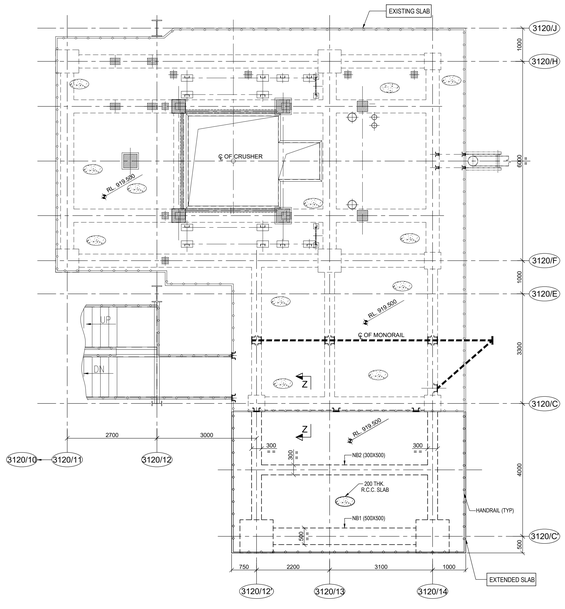


SECTION Y1-Y1
(SCALE-1:50)

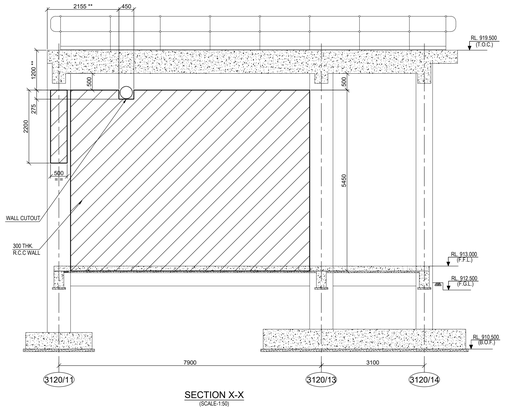


SECTION Y2-Y2
(SCALE-1:50)

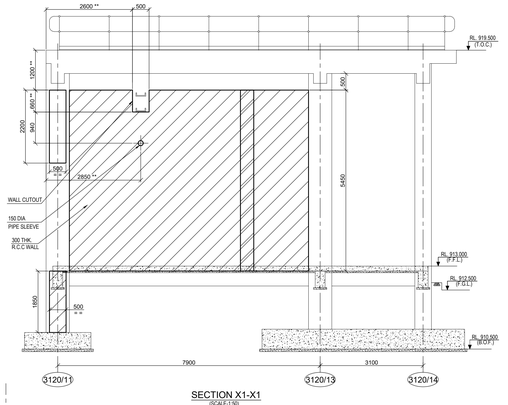
** DIMENSIONS SHALL BE AS PER ACTUAL SITE MEASUREMENT



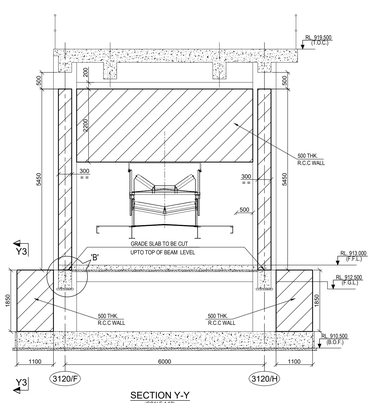
PLAN AT RL 919.500M LEVEL
(SCALE-1:50)



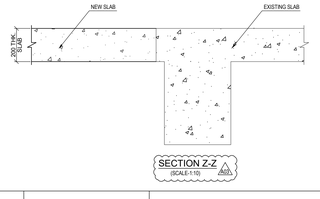
SECTION X-X
(SCALE-1:50)



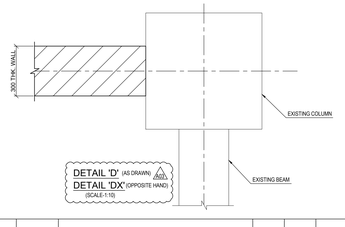
SECTION X1-X1
(SCALE-1:50)



SECTION Y-Y
(SCALE-1:50)



SECTION Z-Z
(SCALE-1:10)



DETAIL 'D' (AS DRAWN)
DETAIL 'DX' (OPPOSITE HAND)
(SCALE-1:10)

LEGENDS :-
RL - REDUCED LEVEL
T.O.C. - TOP OF CONCRETE
F.G.L. - FINISHED GROUND LEVEL
F.F.L. - FINISHED FLOOR LEVEL
B.O.F. - BOTTOM OF FOUNDATION
TYP. - TYPICAL
P.C.C - PLAIN CEMENT CONCRETE
NEW WALL CONSTRUCTION

NOTES :-
1. ALL DIMENSIONS ARE IN MILLIMETRES AND LEVELS ARE IN METRES.
2. GRADE OF CONCRETE SHALL BE C35.
3. NET SAFE BEARING CAPACITY OF SOIL AT BOTTOM OF FOUNDATION CONSIDERED 500KN/SQ. M.
4. FOR DETAIL OF HANDRAIL FIXING REFER STANDARD DRAWING.
5. QUICK SETTING ADMIXTURE SHALL BE USED FOR REDUCING THE SETTING TIME OF THE CONCRETE.
6. DO NOT SCALE, PLEASE ASK.

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3120-EG-CV-DED-1009	SECONDARY CRUSHING R.C DETAILS OF								
	ADDITIONAL FOUNDATION & WALL	A03	14/10/2022	REVISED AS PER EMPLOYER COMMENTS DATED 06/06/2022	SYA	SBY	APR	AG	
MD-216-3120-EG-CV-GAD-1007	SECONDARY CRUSHING DETAILS & SECTIONS	A02	28/04/2022	REVISED AS PER EMPLOYER COMMENTS DATED 20/04/2022	SYA	SBY	APR	AG	
MD-216-3120-EG-CV-GAD-1004	SECONDARY CRUSHING LAYOUT & DETAIL	A01	12/04/2022	ISSUED OF APPROVAL	SYA	SBY	APR	AG	

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LARSEN & TOUBRO DRAWING NUMBER	019045_100_C_G_24675_001
REVISION	2
OUTOTEC DRAWING NUMBER	900707 _ GAU01 _ ZC01 _ MTL01 _ 31393
DOCUMENT ID	OU100750794
REVISION	2
DRAWN	SYA 08/04/2022
CHECKED	SBY 11/04/2022
ENGINEER	APR 08/04/2022
APPROVED	AG 12/04/2022
EMPLOYER	
SCALE	1:50,1:25,1:10
	A0

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
SECONDARY CRUSHING
MODIFICATION OF CRUSHER SUPPORT STRUCTURE

DRAWING NUMBER: MD - 216 - 3120 - EG - CV - GAD - 1011 | REV: A03

9. Appendix 2

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Subrata Dey

From: Joao Mendes <joao.mendes@sandvik.com>
Sent: 25 April 2022 18:43
To: Dhiman Majumdar; Vikash Sinha
Cc: Debiprasad Ghosh; Sonjit Mallick; Abhijit Ghosh; Ranjit Ghosh; Ashwin Prajapati; Mattias Lindh; B.Gangadhara Reddy; Subrata Dey; Jaydip Das; Gopinath T J
Subject: RE: Ma'aden MMGP Crushing System - Vibrations during No Load Trials of CH870i (Vertical Dynamic Load Application)

CAUTION: This is an external email. Do not click links or open attachments unless you recognize the sender and know the content is safe.

Dear Sir,

Please see below in blue the feedback from R&D department.

- Confirmation for the vertical load application philosophy from Sandvik as given in the attachment or their suggested method
  - *The suggested methodology is reasonable.*
- The permissible RMS velocity below isolation pads i.e. top of insert plates.
  - *R&D doesn't have any recommendations on these values. Likely theses values should be stipulated in a relevant standard / design code for the foundation.*

Regards,

João Mendes

From: Dhiman Majumdar <dhimanm@lntecc.com>
Sent: Monday, April 25, 2022 11:33 AM
To: Vikash Sinha <vikash.sinha@sandvik.com>; Joao Mendes <joao.mendes@sandvik.com>
Cc: Debiprasad Ghosh <debiprasadghosh@lntecc.com>; Sonjit Mallick <sonjit@lntecc.com>; Abhijit Ghosh <abhijitg@lntecc.com>; Ranjit Ghosh <ranjitghosh@lntecc.com>; Ashwin Prajapati <ashwin.prajapati@sandvik.com>; Mattias Lindh <mattias.lindh@sandvik.com>; B.Gangadhara Reddy <bgreddy@lntecc.com>; Subrata Dey <subratadey@lntecc.com>; Jaydip Das <Jaydipdas@lntecc.com>; Gopinath T J <gopinathtj@lntecc.com>
Subject: Re: Ma'aden MMGP Crushing System - Vibrations during No Load Trials of CH870i (Vertical Dynamic Load Application)

Dear Sir/ Mr. Joao,

Awaiting response to L&T queries.

Please send it by today as our work cannot continue without Sandvik inputs.

Regards,
Dhiman

From: Dhiman Majumdar
Sent: Friday, April 22, 2022 4:47:16 PM
To: Vikash Sinha <vikash.sinha@sandvik.com>; Joao Mendes <joao.mendes@sandvik.com>
Cc: Debiprasad Ghosh <debiprasadghosh@lntecc.com>; Sonjit Mallick <sonjit@lntecc.com>; Abhijit Ghosh <abhijitg@lntecc.com>; Ranjit Ghosh <ranjitghosh@lntecc.com>; Ashwin Prajapati <ashwin.prajapati@sandvik.com>; Mattias Lindh <mattias.lindh@sandvik.com>; B.Gangadhara Reddy <bgreddy@lntecc.com>; Subrata Dey <subratadey@lntecc.com>; Jaydip Das <Jaydipdas@lntecc.com>; Gopinath T J <gopinathtj@lntecc.com>
Subject: RE: Ma'aden MMGP Crushing System - Vibrations during No Load Trials of CH870i (Vertical Dynamic Load Application)

Dear Joao,

Here is the load application philosophy for both vertical and horizontal direction. Bechtel has confirmed the horizontal load application philosophy and we need the following from Sandvik:

- Confirmation for the vertical load application philosophy from Sandvik as given in the attachment or their suggested method
- The permissible RMS velocity below isolation pads i.e. top of insert plates.

Regards,
Dhiman

-----Original Appointment-----
From: Dhiman Majumdar
Sent: 22 April 2022 12:17
To: Dhiman Majumdar; Vikash Sinha; Joao Mendes; Ashwin Prajapati; Mattias Lindh; B.Gangadhara Reddy; Subrata Dey; Jaydip Das; Gopinath T J
Cc: Debiprasad Ghosh; Sonjit Mallick; Abhijit Ghosh; Ranjit Ghosh
Subject: Ma'aden MMGP Crushing System - Vibrations during No Load Trials of CH870i (Vertical Dynamic Load Application)
When: 22 April 2022 14:00-14:30 (UTC+01:00) Brussels, Copenhagen, Madrid, Paris.
Where: Microsoft Teams Meeting

Dear Sirs,

Please find the invitation for the discussion.

Kindly share it with colleagues who are required to attend.

Microsoft Teams meeting

Join on your computer or mobile app
Click here to join the meeting

Join with a video conferencing device
748619195@t.plcm.vc
Video Conference ID: 134 021 397 7
Alternate VTC instructions

Learn More | Meeting options

Regards,
Dhiman

Sensitivity: LNT Construction Internal Use

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Sensitivity: LNT Construction Internal Use

10. Comment Resolution Sheet

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BECHTEL

Mansourah Massarah Gold Project
Comment Resolution Log Sheet / Compliance Report

Outotec
LARSEN & TOUBRO

PROJECT: MANSOURAH MASSARAH GOLD PROJECT **PROJECT NO.: 216**

STAGE: 4 - PROJECT EXECUTION. **REVIEW LOG DOC. NO. & REV.:**

ACONEX REF. NO. & DATE: BECHTEL-WTRAN-015620, 21 October 2022

STATUS CODE: A - APPROVED, C - APPROVED WITH COMMENTS, R - REVISE & RESUBMIT & O - OTHERS (INFORMATION, REFERENCE, REVIEWED)

DOCUMENT NUMBER	Area / Title	Comment No.	PMC/PMT COMMENT DATE: 21 October 2022	Comment by	EPC Response DATE: 01 February 2024	Status	Remarks
			General Comments (cc)				
MD-216-3120-EG-CV-CAL-1007-A03	3120	1	Static analysis of the Secondary Crusher support structure was carried out during the execution stage along with modal analysis to check the frequency separation from the exciting frequency of the cone crusher. Based on the investigation, the causes of excessive vibration felt on the support structure are: · Misinterpretation of the forced vibration nature of the equipment, · Incompliance with project requirements regarding being away from the range of excitation frequencies and vibration velocity amplitude limit; and · Forced vibration analysis of the structure to determine the vibration velocity was not performed earlier.	Mr. Hong Su	Noted.	Closed	
MD-216-3120-EG-CV-CAL-1007-A03	3120	2	Please indicate heights and thicknesses of all new walls.	Mr. Hong Su	Noted.	Closed	